

MAGNETOM

Function Description System

Host / Imager

Document Version

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1	Introduction	5
1.1	General	5
1.2	Overview	6
1.3	Host (MRC) Configuration	7
1.3.1	Function	7
1.4	Host (MRSC) Configuration	8
1.4.1	Function	8
2	Host CELSIUS R630/R640 for MRC and MRSC	10
2.1	Overview of Host CELSIUS R630	10
2.1.1	General	10
2.2	Motherboard of Host CELSIUS R630	13
2.2.1	Function	13
2.3	Interface Boards of Host CELSIUS R630	16
2.3.1	Function	16
2.4	Drives of Host CELSIUS R630	18
2.4.1	Function	18
2.5	Devices of Host CELSIUS R630	20
2.5.1	Function	20
2.6	Host CELSIUS R640 for MRC and MRSC	22
2.6.1	Overview of Host CELSIUS R640	22
2.6.2	Motherboard of Host CELSIUS R640	22
2.6.3	Interface Boards of Host CELSIUS R640	23
2.6.4	Drives of Host CELSIUS R640	24
2.6.5	Devices of Host CELSIUS R640	24
3	Imager MR 005 Step2 (Dual Opteron)	26
3.1	Overview of Imager MR 005 Step2 (Dual Opteron)	26
3.2	Motherboard of Imager MR 005 Step2 (Dual Opteron)	32
3.2.1	Function	32
3.3	Interface Boards of Imager MR 005 Step2 (Dual Opteron)	36
3.3.1	Function	36
3.4	Drives of Imager MR 005 Step2 (Dual Opteron)	38
3.4.1	Function	38
3.5	PCI_RX Boards at Imager MR 005 Step2 (Dual Opteron)	40
3.5.1	Function	40
4	Imager MR 005 Step2 Standard (Dual Opteron)	42
4.1	Overview of Imager MR 005 Step2 Standard (Dual Opteron)	42
4.2	Motherboard of Imager MR 005 Step2 Standard (Dual Opteron)	47
4.2.1	Function	47
4.3	Interface Boards of Imager MR 005 Step2 Standard (Dual Opteron)	50
4.3.1	Function	50
4.4	Drives of Imager MR 005 Step2 Standard (Dual Opteron)	51

4.4.1	Function	51
4.5	PCI_RX16 Boards at Imager MR 005 Step2 Standard (Standard Opteron) ..	52
4.5.1	Function	52
5	Imager MR 005 Step2 High End (Quad Opteron)	53
5.1	Overview of Imager MR 005 Step2 High End (Quad Opteron)	53
5.2	Motherboard of Imager MR 005 Step2 High End (Quad Opteron)	58
5.2.1	Function	58
5.3	Interface Boards of Imager MR 005 Step2 High End (Quad Opteron)	60
5.3.1	Function	60
5.4	Drives of Imager MR 005 Step2 High End (Quad Opteron)	61
5.4.1	Function	61
5.5	PCI_RX16 Boards at Imager MR 005 Step2 High End (Quad Opteron)	62
5.5.1	Function	62
6	Imager MR 005 Step3 Standard	63
6.1	Overview of Imager MR 005 Step3 Standard	63
6.2	Motherboard of Imager MR 005 Step3 Standard	65
6.2.1	Function	65
6.3	Interface Boards of Imager MR 005 Step3 Standard	66
6.3.1	Function	66
6.4	Drives of Imager MR 005 Step3 Standard	67
6.4.1	Function	67
6.5	PCI_RX16 Boards at Imager MR 005 Step3 Standard	68
6.5.1	Function	68
7	Changes to previous version	69
8	List of Hazard IDs	70

1.1 General

The Host computer concept for Tim systems (e.g. MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) systems is the same as used for all syngo MR systems, i.e. the Host is based on a PC.

This is true for main consoles (MRC) and for optional satellite consoles (MRSC).

The Imager for Tim systems (MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) is still based on a PC system, however, the concept for receiving and demodulating data has changed.

1.2 Overview

Although the Host and Imager components are individual components, they have been combined here in Part 4 since they work very closely together.

Before you continue, a few words on the designations used throughout the Host description may be in order to avoid unnecessary confusion.

The term Host refers simply to a computing element which is used to control or act as the "host" of a system. The "host" of our MR system is termed "MR Console", or MRC for short. In the very first product lines the Host computers consisted of separate mini-frame or workstation type computing elements quite large in size and placed in a technical room some distance away from the user. The keyboard, mouse and other user input devices were then physically mounted onto a nice table, or console, and connected to the computer over lots of cables. Hence the term Console. Of course today's computers are compact and house all the mass media devices and provide the connections for the external user interface devices (mouse, keyboard, headphones, etc.) so that the user sits very close to the unit. The MR consoles of today are very high-quality desks with some type of noise enclosure to "hide" the Host hardware.

In addition to the MRC, an optional satellite MR console, or MRSC is available. Although practically the same as the MRC, it is only intended for post-processing images. For this reason, it may not contain some of the hardware components found in the MRC. Also, the installed and configured software will be different, since the MRSC doesn't require all that system control software! A description of the main differences is provided on the next page.

1.3 Host (MRC) Configuration

1.3.1 Function

The major tasks of the Host of the MRC are:

- User Interface (syngo) - provide the hardware and software for the user interface
- Convert sequence measurement parameters from the user into system-specific commands and pass these on to the Advanced Measurement Control (AMC) of the MR system
- Image post-processing and image display functionalities
- Archiving (mass storage, filming)

The MRC is connected to the MR system via an ethernet LAN connection. Since it is to be connected to two other devices, a 4-port switch is used. This network connection will be referred to as the "internal network".

The current Host model is equipped with the following components:

- Computer unit:
 - 3 hard drives, one for the OS and NUMARIS, one for the database and one for the pixel image files
 - 1 CD Burner (CD-R) or DVR-Recorder for image archiving
 - 1 DVD-ROM drive for software updates
 - 1 floppy drive for installing license keys
 - 1 serial port
 - 8 additional serial ports on one MOXA interface card
 - 2 network interfaces: one for system internal communication and one for connection to the customer's LAN
- High-resolution LCD monitor for image and dialog display
- Keyboard, mouse and mouse pad

1.4 Host (MRSC) Configuration

1.4.1 Function

- Image viewing
- Post-processing
- Archiving images

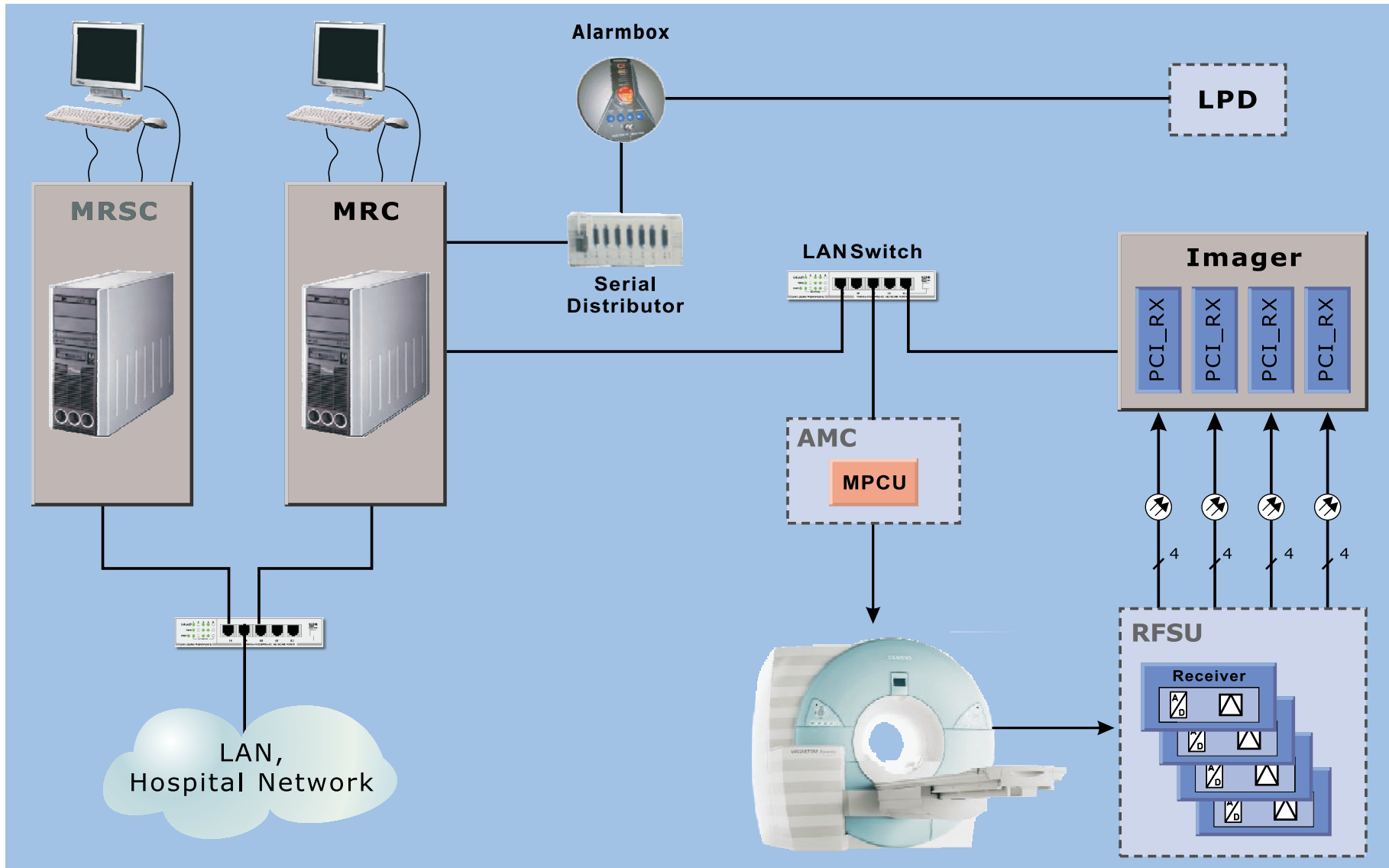
The MR satellite console (MRSC) is an option for

The MRSC Host connects to the Host MRC over an additional Ethernet switch, delivered with the MRSC option. This switch is required to decouple the large volume of image data transfers from the customer's LAN. This network connection will be referred to as the "outside network", or "Customer network".

The configuration of the MRSC computer is the same as the MRC Host with the following exceptions:

- NO pixel data hard drive (database and pixel data on one hard disk only)
- 1 network interface to communicate to the local area network (LAN) and the main console (MRC)
- NO MOXA interface card.

Fig. 1: Host / Imager Block Diagram



2.1 Overview of Host CELSIUS R630

2.1.1 General

The following figure shows the components of the MR Console (MRC) and MR Satellite Console (MRSC):

- High resolution **LCD Display**: to look at
- **keyboard** and **mouse**: Input devices for entering text and pressing buttons
- Host (Intel based PC) - basic model identical for both MRC and MRSC, however, the configuration is different.

Since developments in the computer industry are advancing rapidly, this description applies only to the current configuration of the model implemented at the time of creation or modification of this document. Future versions will of course have faster processors, more memory, larger disk capacity, etc.

Fig. 2: Snapshot MR Console



2.1.1.1 Host sub-sections

The following figure shows a breakdown of the Host into its major subsections:

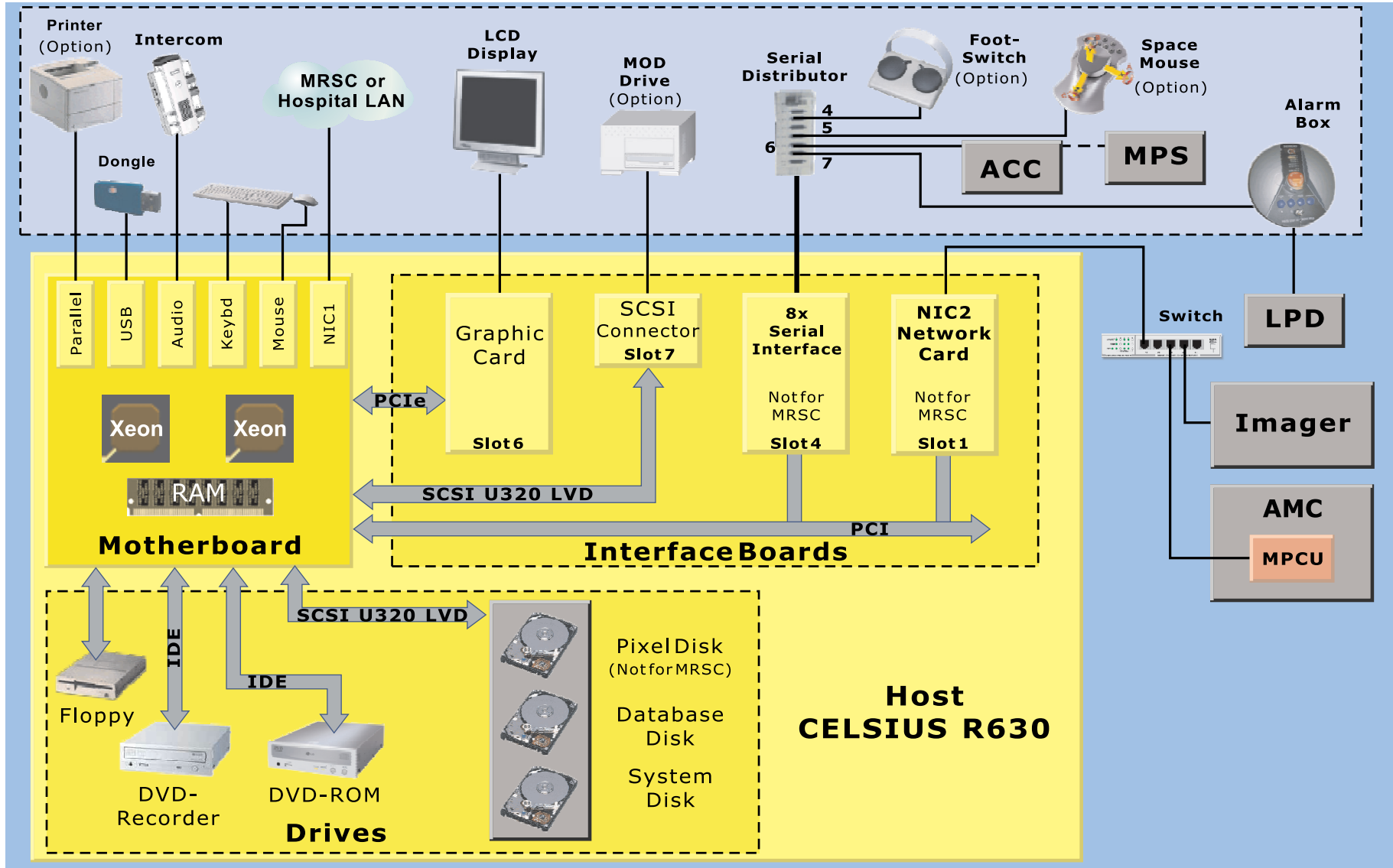
- Motherboard
- Drives
- Interface Cards
- Connected Devices.

The block diagram shows you in detail all connections established outside the Host tower. The internal connections between motherboard and interfaces and drives will be shown in greater detail in the following pages.

The connections, interfaces, drives and devices not available for MRSC Hosts are marked with "Not for MRSC".

The Avanto MR systems use a release of the syngo software which uses Windows XP as the operating system.

Fig. 3: Host CELSIUS R630 for MRC



2.2 Motherboard of Host CELSIUS R630

2.2.1 Function

There are four different areas on the motherboard used for MRC Hosts and for MRSC Hosts as well. These areas are shown in the following figure:

- CPU and RAM
- Connections in back
- Connections in front
- Internal interfaces (controllers and adapters)

2.2.1.1 CPU and RAM

The CPU and RAM area of the dual processor motherboard is equipped with:

- 2x 3.6 GHz Intel Xeon **CPUs** with 1 MB second level cache,
- 2GB DDR SDRAM (3GB with syngo MR B13); 4 available DIMM slots with 2 x 1GB.

2.2.1.2 Connections in back

The on-board interfaces that can be reached from the back of the Host are:

- Parallel port (DB-25 pin) - for optional printer
- 4x USB 2.0 ports - the right port in back is used for connecting the dongle
- 1x IEEE 1394, Firewire (not used)
- Audio - line input, headphone output, microphone input (Codec97 compatible) and digital output S/PDIF
- 2x PS/2 ports - one keyboard, one mouse
- On-board Broadcom BCM5751 Gbit LAN controller - 1000 Mbit communication to the MRSC or to the LAN or hospital network
- Serial port (DB-9 pin) - not used
- VGA interface
- MOXA interface card

2.2.1.3 Connections in front

The on-board interfaces that can be reached from the front of the Host are:

- 2x USB 2.0 ports (not used)
- 1x IEEE 1394, Firewire (not used)
- On-board audio interface provides an additional headphone output and a microphone input in front

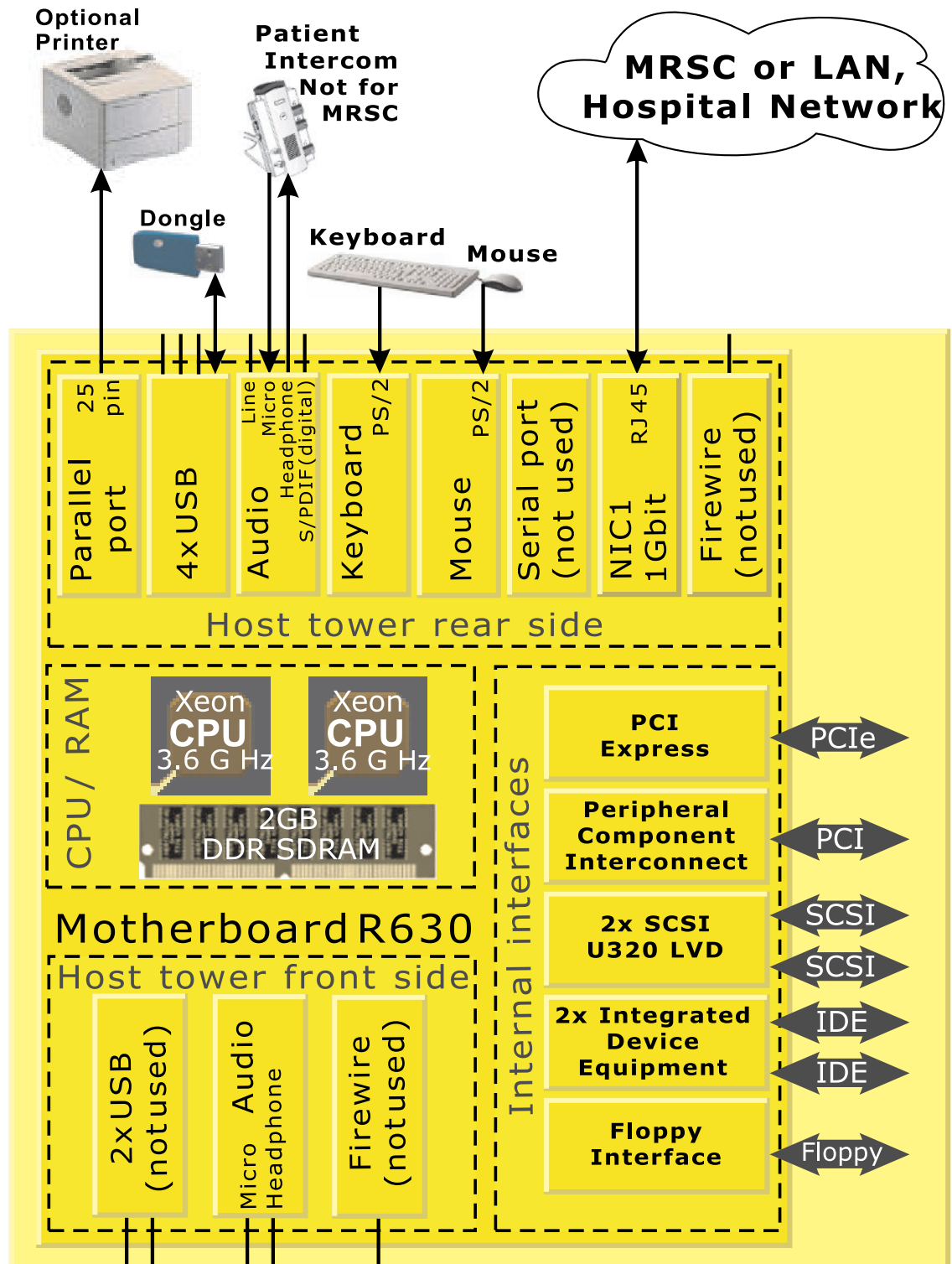
2.2.1.4 Internal interfaces

The following adapters and controllers are implemented on the motherboard:

- PCIe graphic card (currently Nvidia FX3400)

- On-board SCSI U320 LVD controller for all hard drives
- Second on-board SCSI U320 LVD controller for external MOD drive. The cabling runs internally and the connector is located at the cover of slot 7. This SCSI bus is terminated at the connector of slot 7.
- 2x IDE (Integrated Device Equipment) bus connections.
 - One for the DVD recorder
 - One for the DVD-ROM
- On-board Floppy controller interface

Fig. 4: Host CELSIUS R630 Motherboard Overview



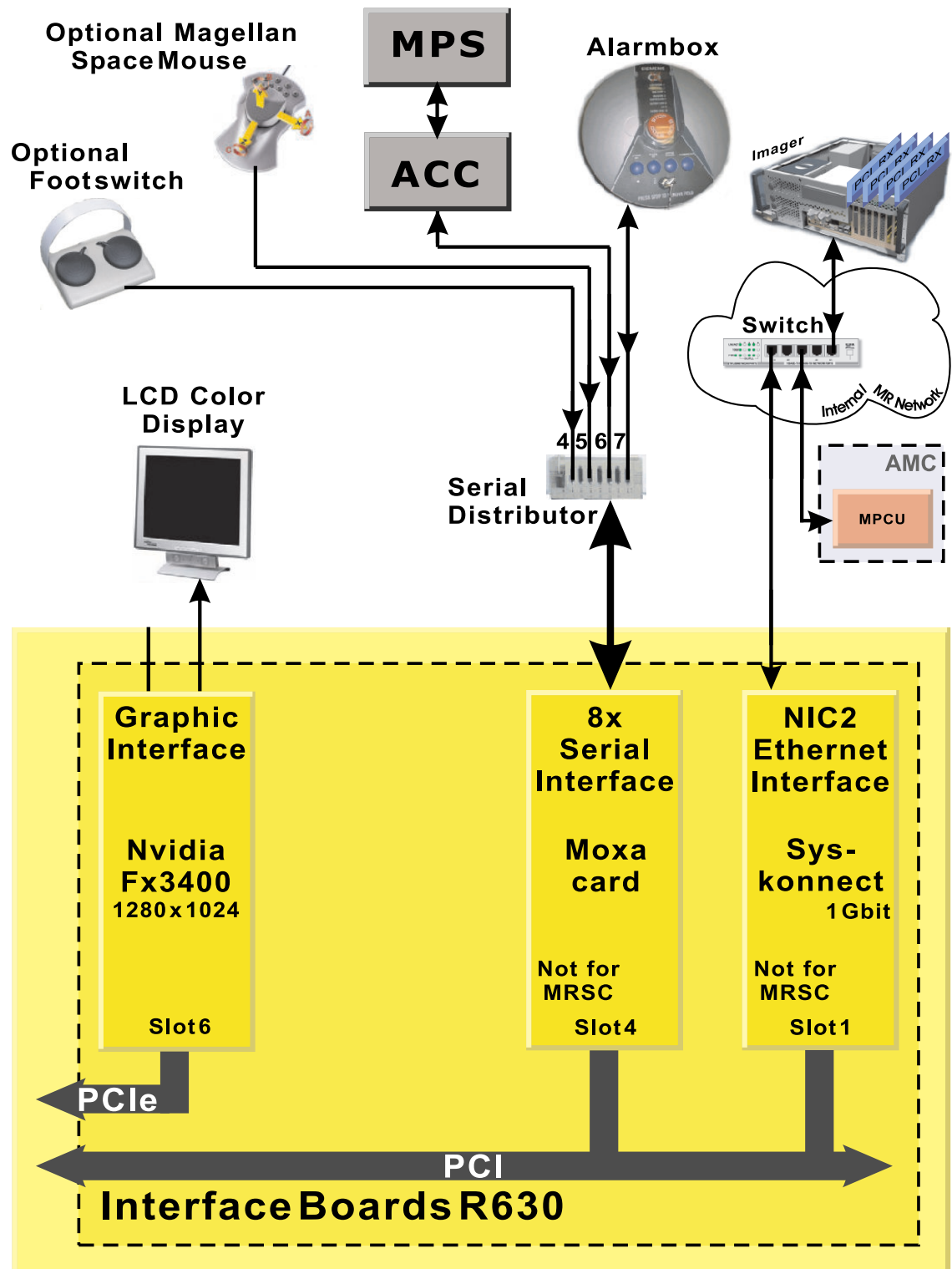
2.3 Interface Boards of Host CELSIUS R630

2.3.1 Function

There are two PCI adapter cards and one PCI Express (PCIe) card installed in the MRC Host and just one PCI Express (PCIe) card is installed in the MRSC Host. These cards can be accessed in back of the Host at different slots as shown in the following figure:

- The graphic card is a Nvidia FX3400 and placed in the PCIe slot. The graphic card hardware provides two digital video outputs - the software just provides one output. The resolution is set to 1280 x 1024, the resolution of the syngo software. The graphic card is located at slot 6 in back of the Host tower.
- The 8x serial interface board (MOXA card) is the PCI card located at slot 4 and is used to provide eight additional serial input / output ports. The MOXA card connects via cable to an external serial distributor box mounted in back of the operator's console which provides 8x 25-pin serial ports:
 - port 7 - the Alarmbox
 - port 6 - the Magnet Power Supply
 - port 5 - the optional Magellan Space Mouse
 - port 4 - the Footswitch (part of the in-room option)
 - The rest of the ports are available for future use. The 8x serial interface board, the serial distributor and the cable between are available for the MRC Hosts only. As a result, these components can be connected only to the MRC Host.
- The Network card is a Syskonnect Gbit Ethernet card located at slot 1. It allows you to run networking communication with one Gbit per second transfer rate.

Fig. 5: Host CELSIUS R630 Interface Boards Overview



2.4 Drives of Host CELSIUS R630

2.4.1 Function

There are four on-board drive controllers connected to the drives shown in the following figure, the hard drive configuration data are shown in the table below:

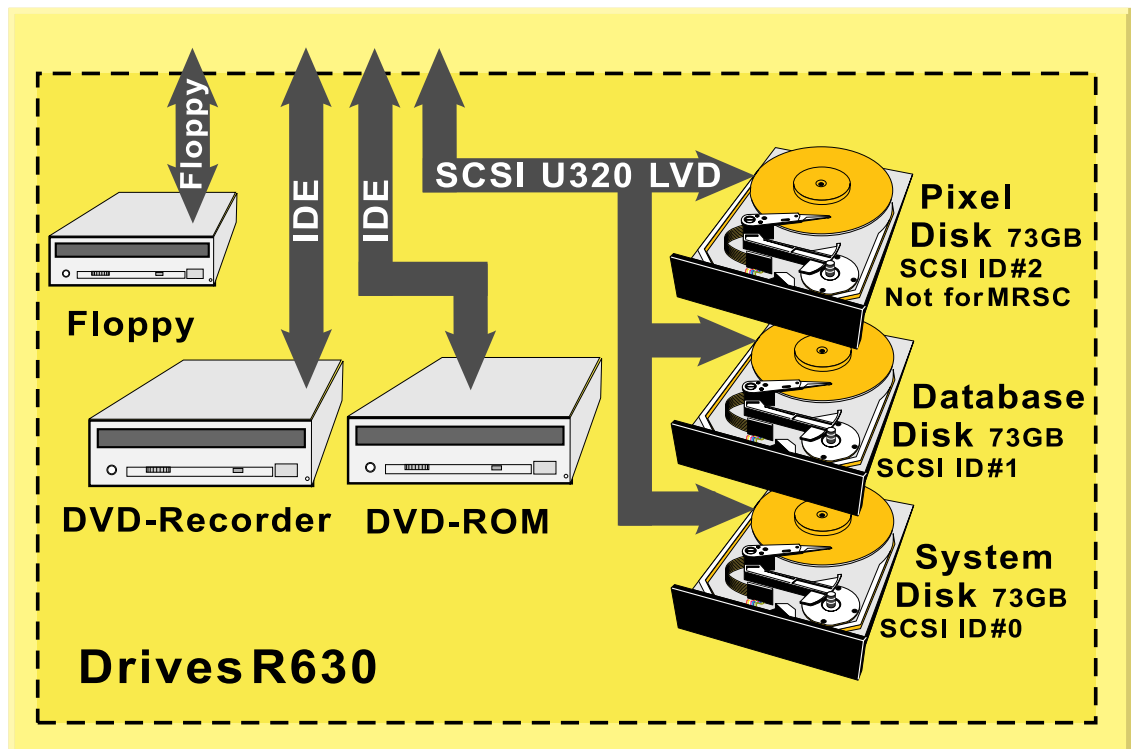
- The Floppy drive is connected to the internal Floppy controller on the motherboard. The Floppy drive is needed to install new licenses when purchasing additional options or to install licenses after a software reload. Exchanging the dongle or the Host will require new licenses to be issued and installed.
- The DVD recorder (DVD +R/RW) is an IDE device and hangs on the internal IDE controller interface. The DVD recorder drive is intended for image archiving and software backups.
- The DVD-ROM drive is also an IDE device connected to the internal IDE controller interface. The DVD-ROM drive is intended for software installation, reloads, and updates.
- The hard drives (HDD) are connected to the on-board SCSI controller. The SCSI controller used in the current model is the 16 bit SCSI U 320 LVD standard. "U 320" refers to the Ultra SCSI standard which has a transfer rate of 320 MByte per second. LVD is an abbreviation for "Low Voltage Differential SCSI".
- The MOD option is available for both the MRC and MRSC Hosts and is connected to the second on-board SCSI U320 LVD controller.

Tab. 1 Hard Drive Configuration

Drive	SCSI ID	Size	Use
C	0	73 GB	System Disk
D	1	73 GB	Database (Image headers)
E	2	73 GB	Pixel data (Images)

External drives such as the optional MOD drive are shown in the next figure.

Fig. 6: Host CELSIUS R630 Drives Overview



2.5 Devices of Host CELSIUS R630

2.5.1 Function

There are several external devices connected to the MRC and MRSC Hosts. An overview of external devices is shown in the following figure.

- Network switch for the internal MR network
- LCD Color Display
- Optional MOD drive
- Optional printer
- Optional network switch for MRC, MRSC and LAN connections
- Dongle
- Patient Intercom (via central unit)
- Keyboard
- Mouse
- Serial Distributor / Alarmbox
- Magellan Space Mouse (option)
- Footswitch (option)

Let's discuss those external devices at greater length:

- The network switch for the internal MR network: This switch connects NIC2 of the MRC Host, the Ethernet controller (NIC) of the MPCU and the Ethernet controller (NIC) of the Imager to build up the internal MR network for internal communications between the MRC Host, the Imager and the Advanced Measurement Control (AMC) together with all peripheral MR units. Of course, there is no internal switch and no internal MR network for the MRSC Hosts. The communication running on the internal MR network uses a transfer rate of 1 Gbit per second. The NICs and the switch are connected with twisted pair straight-through cables (patch cables). This switch uses 4 network ports.
- A 1280x1024 18" LCD Color Display is used at the MRC and MRSC Hosts and driven by a Nvidia FX3400 graphic card (slot 6).
- The optional MOD, Pioneer 1.7 GB, is connected to the second on-board SCSI bus (U320 LVD) of the Host running SCSI-2 communication. This allows importing of NUMARIS/3 and NUMARIS/3.5 images into the syngo MR database. The software defines drive letter F: for that drive. This option is available for MRC and MRSC Hosts. The SCSI connector is located in back of the Host at the cover of slot 7.
- An optional printer can be connected via a parallel port in back of the MRC and MRSC Host (25 pin Cannon connector).
- An optional network switch is delivered with the MRSC option to connect to the MRC. It connects the MRC Host via Ethernet controller NIC1, the MRSC Host and the Hospital Network or LAN. If an MRSC is installed, a lot of communication has to be handled between the MRC Host and the MRSC Host. This switch allows internal communication to be independent of data transfer or network problems at the hospital network. The communication running on the optional network switch uses a transfer rate of 1 Gbit

per second. The NICs and the switch are connected with twisted pair straight-through cables (patch cables). This switch uses 4 network ports.

- **The Dongle:** Since Avanto MR systems are running Windows XP, professional USB dongles are used and checked whenever you start the service software or any features which require a license key. The USB dongle is connected at the right USB port of the four USB 2.0 ports in back of the Host. A dongle has to be used for each console, i.e. you need a dongle for the MRC Host and a separate dongle for the MRSC Host. The dongle IDs for the USB dongles start with Flex-Id9.
- **The Patient Intercom:** The headphone output and the microphone input of the Audio card at the MRC Host are connected to the Patient Intercom to play music in the RF room for the patients' comfort, for using the voice output option and for giving commands to the patient before and during the measurement. There is, of course, no patient intercom at the satellite consoles (MRSC).
- **The keyboard used** is basically a standard Windows PC keyboard, but the numerical keys use the syngo symbols. We deliver country-specific keyboards to the different countries. The same keyboard is used for main and satellite consoles.
- **The mouse we use** is a standard PC Mouse. It is a mechanical one with 3 keys. The same mouse is used for main and satellite consoles.
- **The serial distributor** is available at the MRC Hosts only. It just provides a 25 pin connector for each of the eight serial interfaces found at the 8x serial interface reachable via slot 4 of the MRC Host.
- **The alarm box** of the MR scanner is always connected to serial port 7 of the serial distributor.
- **The optional Magellan Space Mouse:** The Magellan Space Mouse is used for real-time interactive imaging. If available, it is connected to serial port 5 of the serial distributor.
- **The optional Footswitch:** The footswitch located in the RF room is used to start protocols which were already prepared at the MRC. It is also used to stop running protocols directly in the RF room.
- **The Magnet Power Supply MPS :** The MPS is used for ramping up and down the magnet system. It is connected to the Host via serial port 6 of the serial distributor.

2.6 Host CELSIUS R640 for MRC and MRSC

2.6.1 Overview of Host CELSIUS R640

2.6.1.1 General

The components of the MR Console (syngo Acquisition Workplace, MRC) and MR Satellite Console (syngo MR Workplace, MRSC) are:

- High resolution **LCD Display**: to look at
- **keyboard** and **mouse**: Input devices for entering text and pressing buttons
- Host (Intel based PC) - basic model identical for both MRC and MRSC, however, the configuration is different.

Since developments in the computer industry are advancing rapidly, this description applies only to the current configuration of the model implemented at the time of creation or modification of this document. Future versions will of course have faster processors, more memory, larger disk capacity, etc.

Host sub-sections

The major subsections of the Host are:

- Motherboard
- Drives
- Interface Cards
- Connected Devices.

2.6.2 Motherboard of Host CELSIUS R640

2.6.2.1 Function

There are four different areas on the motherboard used for MRC Hosts and for MRSC Hosts as well:

- CPU and RAM
- Connections in back
- Connections in front
- Internal interfaces (controllers and adapters)

CPU and RAM

The CPU and RAM area of the dual processor motherboard is equipped with:

- 2x 2.6 GHz Intel Woodcrest **CPUs**,
- 4GB DDR RAM; 4 available DIMM slots with 4x 1GB.

Connections in back

The on-board interfaces that can be reached from the back of the Host are:

- Parallel port (DB-25 pin) - for optional printer
- 4x USB 2.0 ports - one port in back is used for connecting the dongle
- 1x IEEE 1394, Firewire (not used)
- Audio - digital output S/PDIF
- 2x PS/2 ports - one keyboard, one mouse
- 2x On-board 1 Gbit LAN controller
- Serial port (DB-9 pin) - not used
- VGA interface
- MOXA interface card

Connections in front

The on-board interfaces that can be reached from the front of the Host are:

- 2x USB 2.0 ports (not used)
- 1x Firewire (not used)
- On-board audio interface provides an additional headphone output and a microphone input in front

Internal interfaces

The following adapters and controllers are implemented on the motherboard:

- PCIe graphic card (currently Nvidia FX3500)
- On-board SCSI U320 LVD controller for all hard drives
- 2x IDE (Integrated Device Equipment) bus connections.
 - One for the DVD recorder
 - One for the DVD-ROM
- On-board Floppy controller interface

2.6.3 Interface Boards of Host CELSIUS R640

2.6.3.1 Function

There are two PCI adapter cards and one PCI Express (PCIe) card installed in the MRC Host and just one PCI Express (PCIe) card is installed in the MRSC Host. These cards can be accessed in back of the Host at different slots as shown in the following figure:

- The graphic card is a Nvidia FX3500 and placed in the PCIe slot. The graphic card hardware provides two digital video outputs - the software just provides one output. The resolution is set to 1280 x 1024, the resolution of the syngo software. The graphic card is located at slot 6 in back of the Host tower.

- The 8x serial interface board (MOXA card) is the PCI card located at slot 4 and is used to provide eight additional serial input / output ports. The MOXA card connects via cable to an external serial distributor box mounted in back of the operator's console which provides 8x 25-pin serial ports:
 - port 7 - the Alarmbox
 - port 6 - the Magnet Power Supply
 - port 5 - the optional Magellan Space Mouse
 - port 4 - the Footswitch (part of the in-room option)
- The rest of the ports are available for future use. The 8x serial interface board, the serial distributor and the cable between are available for the MRC Hosts only. As a result, these components can be connected only to the MRC Host.

2.6.4 Drives of Host CELSIUS R640

2.6.4.1 Function

There are four on-board drive controllers connected to the drives, the hard drive configuration data are shown in the table below:

- The Floppy drive is connected to the internal Floppy controller on the motherboard. The Floppy drive is needed to install new licenses when purchasing additional options or to install licenses after a software reload. Exchanging the dongle or the Host will require new licenses to be issued and installed.
- The DVD recorder (DVD +-R/RW) is an IDE device and hangs on the internal IDE controller interface. The DVD recorder drive is intended for image archiving and software backups.
- The DVD-ROM drive is also an IDE device connected to the internal IDE controller interface. The DVD-ROM drive is intended for software installation, reloads, and updates.
- The hard drives (HDD) are connected to the on-board SCSI controller.

Tab. 2 Hard Drive Configuration

Drive	SCSI ID	Size	Use
C	0	73 GB	System Disk
D	1	73 GB	Database (Image headers)
E	2	73 GB	Pixel data (Images)

2.6.5 Devices of Host CELSIUS R640

2.6.5.1 Function

There are several external devices connected to the MRC and MRSC Hosts. An overview of external devices is shown in the following figure.

- Network switch for the internal MR network
- LCD Color Display

- Optional printer
- Optional network switch for MRC, MRSC and LAN connections
- Dongle
- Patient Intercom (via central unit)
- Keyboard
- Mouse
- Serial Distributor / Alarmbox
- Magellan Space Mouse (option)
- Footswitch (option)

3.1 Overview of Imager MR 005 Step2 (Dual Opteron)

The Imager MR 005 Step2 (MRIR) a 19 inch rack-mounted industrial PC is located in the ACC cabinet.

This Imager MR 005 Step2 (MRIR) is used for all Tim systems (f. e. MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) systems.

The MR signal data coming from the RF system and demodulated as well as filtered at the PCI_RX will be temporarily stored on pixel disks. The task of the Imager MR 005 Step2 (MRIR) is to collect MR signals from the RFSU, to calculate the raw data, and to finally perform image reconstruction. The raw data are stored temporarily on the raw data disks. Two AMD processors perform image reconstruction after the measurement is finished. The resulting images are transferred via the internal MR network to the patients' database of the MRC Host.

Raw data will be available for display only when enabling the storage prior to the measurement. The second possibility of displaying raw data is a backward FFT of the images.

The task of the Imager MR 005 Step2 (MRIR) is:

- to receive the measurement data from the RF system
- to calculate the raw data during demodulation
- to reconstruct, to display, and to evaluate MR images

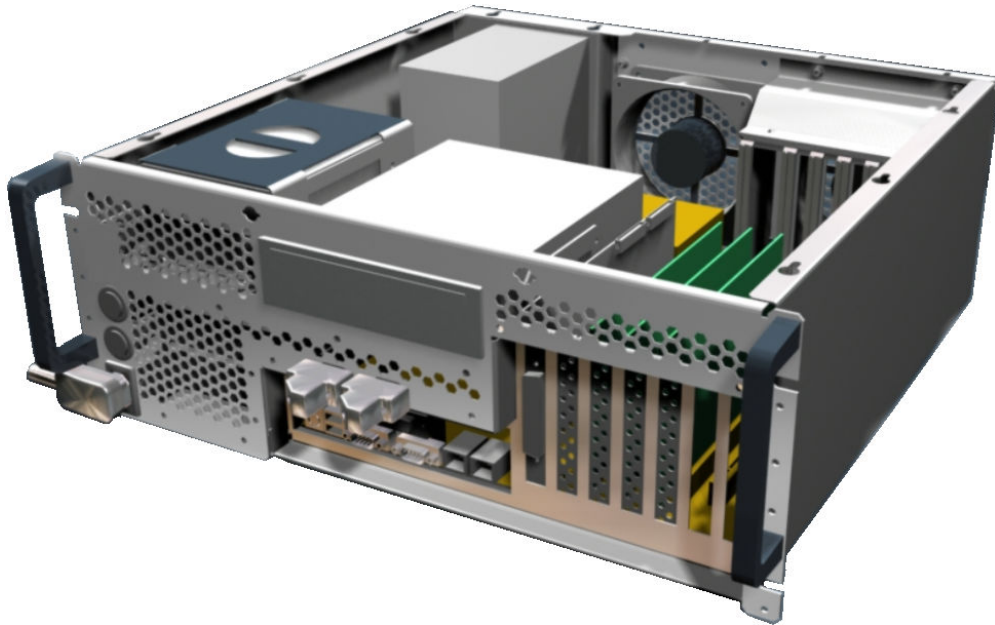
The next figure shows the 19 inch component of an Avanto Imager MR 005 Step2. It includes:

- the motherboard,
- five hard disk drives,
- several standard interfaces,
- and MR-specific interfaces.

The details of the Imager MR 005 Step2 are pretty similar to the details of the Imager MR 005 discussed before. The differences are discussed below.

However, due to rapid developments in the computer world, this description shows the conditions at the time of creation or modification of this document.

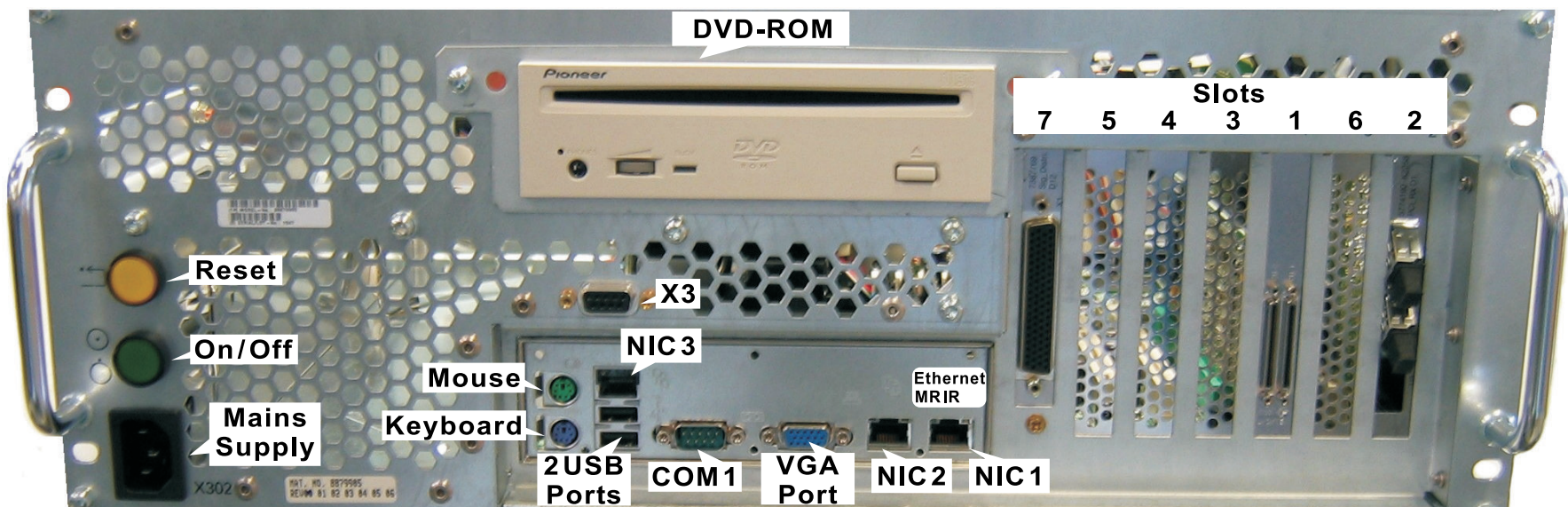
Fig. 7: Imager Unit - MR 005 Step2



A block diagram of the Imager MR 005 Step2 systems used is shown in the following figure. The diagram shows the different areas on the Imager MR 005 Step2 component, such as the motherboard, the drives, and the interface boards.

We have started to use a new syngo software release, running Windows XP as an operating system software on the Imager MR 005 Step2 (for software syngo MR 2004V and syngo MR 2005E). Since syngo MR B13, we run Linux as the operation system software.

Fig. 8: Front view of Imager MR005 Step2 Dual Opteron



Tab. 3 Imager MR005 Dual Opteron Buttons and Indicators

Label	Use
Reset	Reset button
On / Off	On / Off button

Tab. 4 Imager MR005 Dual Opteron Drives

Label	Use
DVD-ROM	DVD-ROM drive for software installation

Tab. 5 Imager MR005 Dual Opteron Front Connectors

Label	Use
Mains Supply	Power supply connection
X3	Imager fan supervision
Mouse	Keyboard and Mouse emulator connection
Keyboard	Keyboard and Mouse emulator connection
2 USB Ports	not used: Two USB ports
COM1	not used: Serial Port
VGA Port	Video output, for troubleshooting only
NIC1	Ethernet controller 1 for internal MR network
NIC2	not used: Ethernet controller 2
NIC3	not used: Ethernet controller 3



Please take a look at the strange order of the slot numbers on the Imager MR 005 Step2. This order is correct!

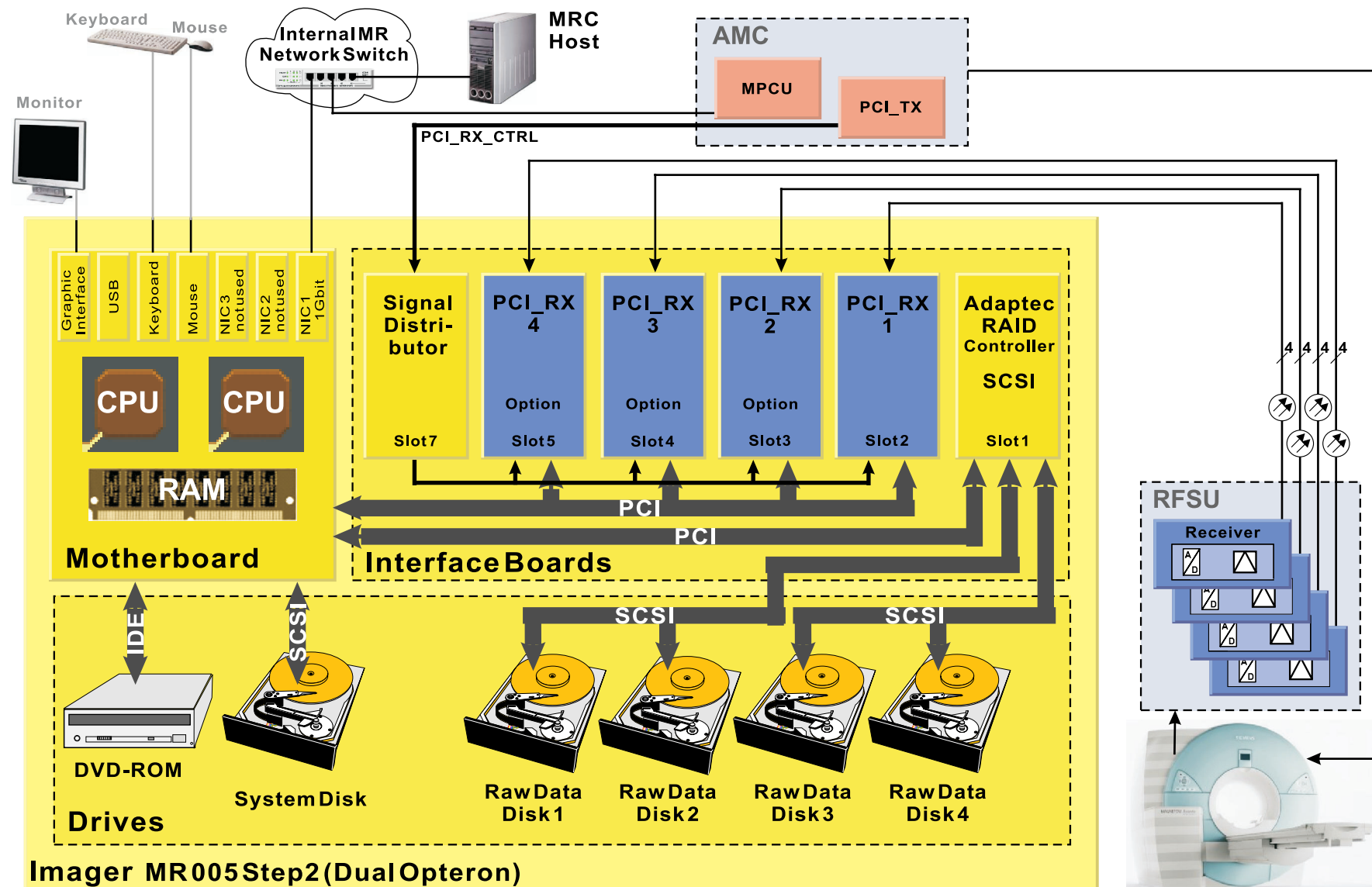
The Imager MR 005 and the Imager MR 005 Step2 use different motherboards with different functionalities of the PCI slots. This causes a different order of the PCI boards (PCI_RX boards and RAID controller). However the labels of all FOCs from the receiver modules (RFSU) to the PCI_RX boards (MRIR) use slot numbers (e.g.: U1 PCI_RX/SLOT2 MRIR/D1). To be able to manufacture and use the same FOCs with the same labels for all imagers, we rearranged the order of the front labels at the Imager MR 005 Step2.

Tab. 6 Imager MR005 Dual Opteron Slots

Label	Use
Slot 2	PCI_RX1 board
Slot 6	not used

Label	Use
Slot 1	Adaptec RAID controller
Slot 3	PCI_RX2 board (optional)
Slot 4	PCI_RX3 board (optional)
Slot 5	PCI_RX4 board (optional)
Slot 7	Signal Distributor

Fig. 9: Imager MR005 Step2 Dual Opteron



3.2 Motherboard of Imager MR 005 Step2 (Dual Opteron)

3.2.1 Function

The Imager MR 005 Step2 motherboard can be broken down into three functional blocks:

- CPU and RAM area
- on-board controllers
- peripheral interfaces (all accessible at the front side of the Imager MR 005 Step2)

3.2.1.1 CPU and RAM

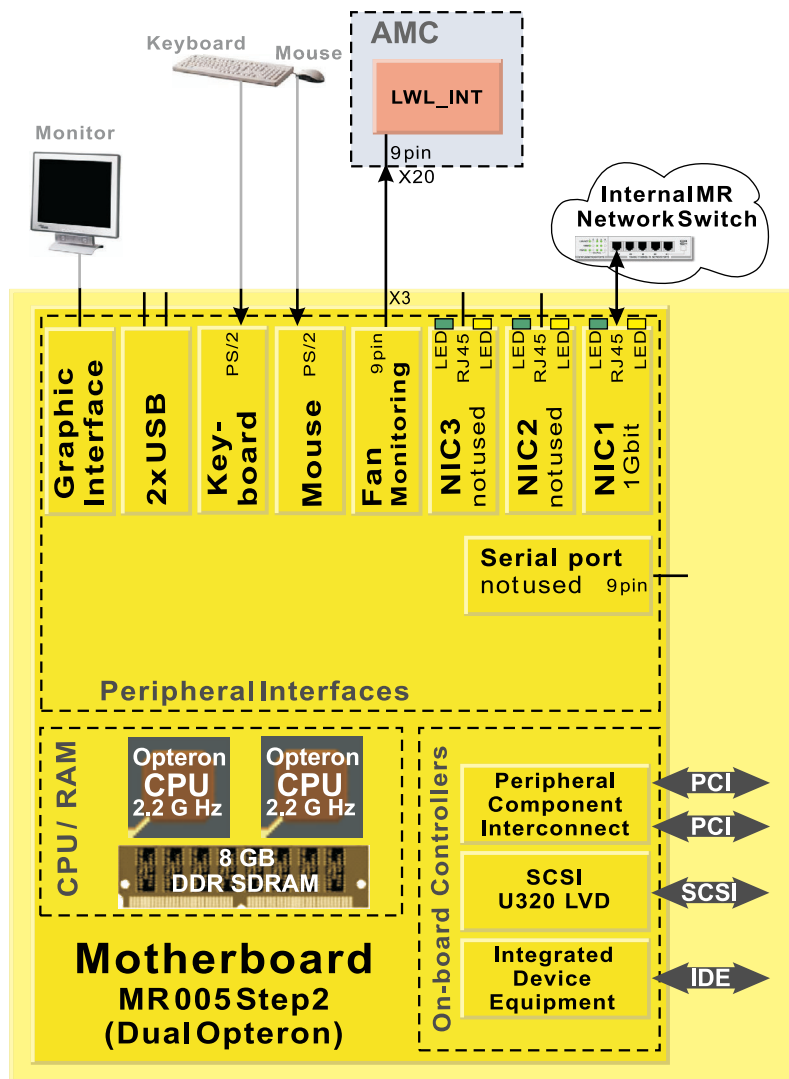
The CPU and RAM area of the dual processor motherboard is equipped with:

- Dual 2.2 GHz AMD Opteron **CPUs** (64 bit CPUs) with 1MB second level cache
- 8 GB DDR SDRAM (4 GB are used with the software syngo MR 2004V and syngo MR 2005E only)

3.2.1.2 On-board Controllers

- up to 4 PCI_RX boards
- 1x PCI SCSI Controller card (Adaptec RAID-capable controller) for control of the four raw data scratch disk
- 1x onboard SCSI controller, an Adaptec AIC-7902 U320-SCSI-Controller for the system disk
- 1x IDE controller connection for the DVD-ROM drive

Fig. 10: Imager MR005 Step2 Motherboard - Overview



3.2.1.3 Peripheral Interfaces

The peripheral interfaces include:

- 2x PS/2 ports, one keyboard, one mouse. The Imager MR 005 Step2 is not equipped with a keyboard or a mouse or a keyboard + mouse emulator.
- 2x USB 1.1 ports - on the front side - not used
- Serial port (DB-9 pin) - not used
- On-board graphic controller (ATI Rage XL) - to connect a monitor, if necessary
- 2x ethernet adapters (Broadcom Gigabit) running at 1 Gigabit (NIC1, NIC2). The first ethernet adapter, NIC 1 (right connector), is used to connect the Imager MR 005 Step2 to the Host and AMC.
- 1x ethernet adapter (Intel 10/100 Megabit), which is not used (NIC3).

Fig. 11: Imager MR005 Step2 Peripheral Interfaces

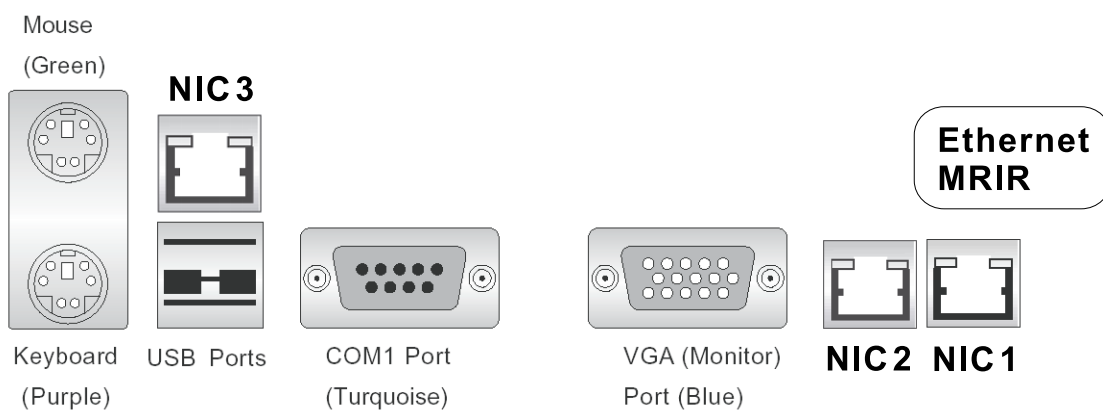
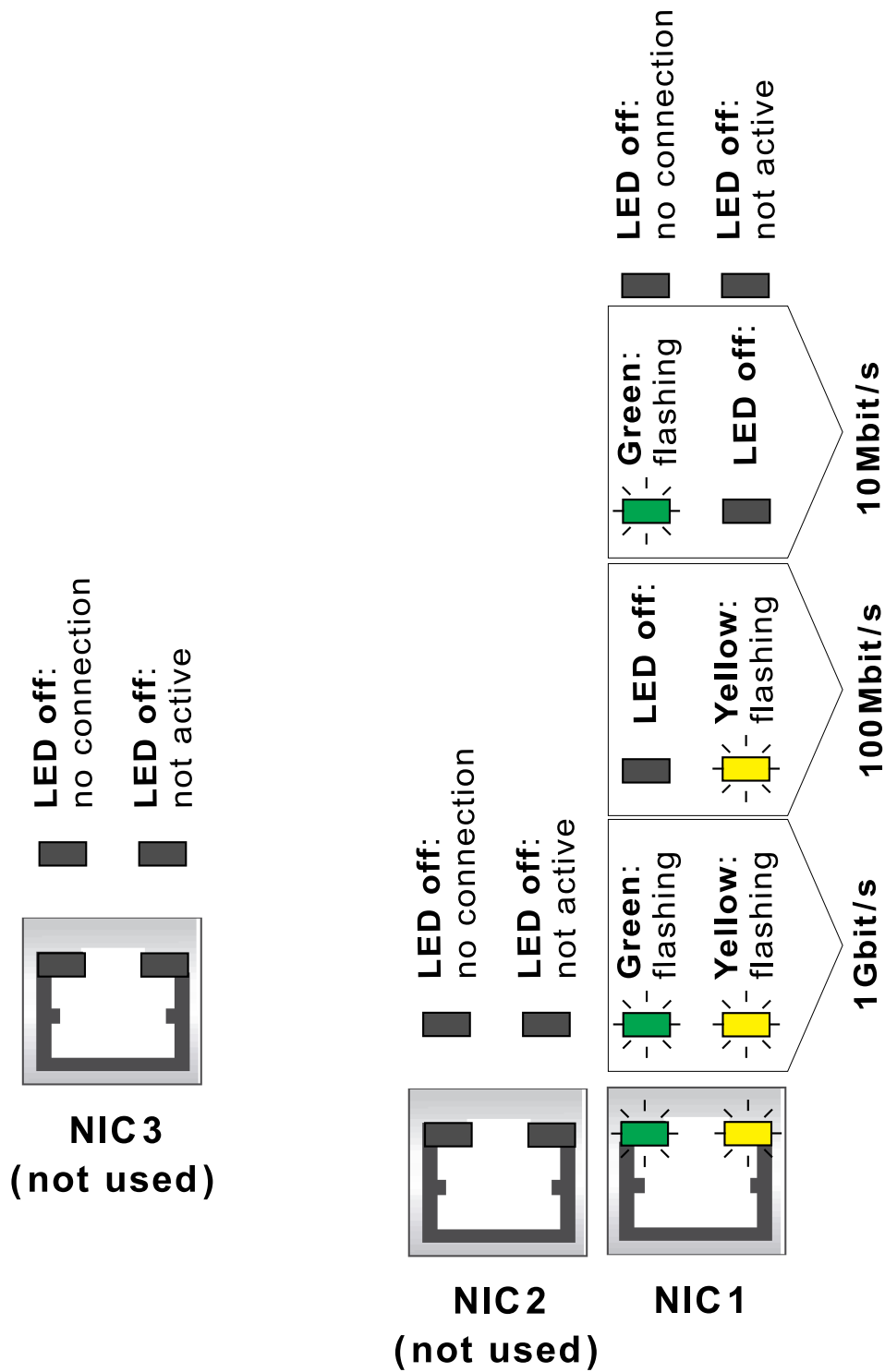


Fig. 12: Imager MR005 Step2 NIC LEDs



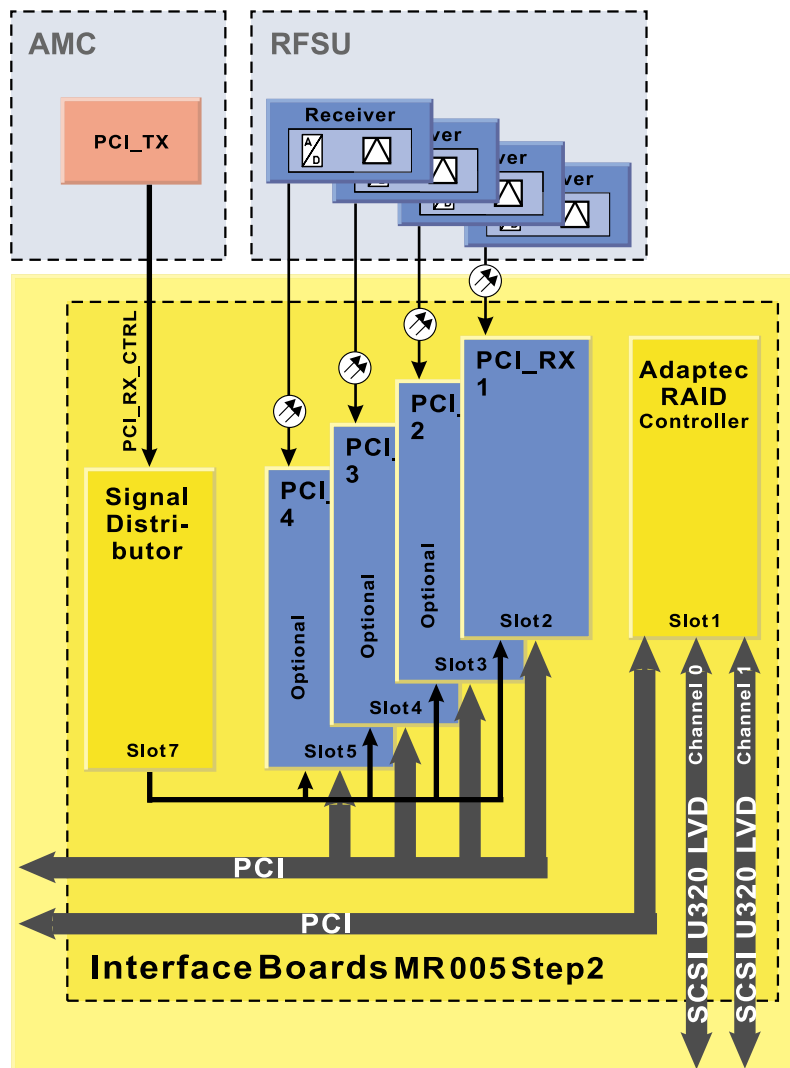
3.3 Interface Boards of Imager MR 005 Step2 (Dual Opteron)

3.3.1 Function

There are different interface boards located at the Imager MR 005 Step2. These interface boards can be accessed at the front side of the Imager MR 005 Step2 at different slots:

- The signal distributor located at slot 7 of the Imager MR 005 Step2. It distributes the control signals from the PCI_TX in the AMC to all available PCI_RX boards.
- One PCI_RX board as a standard, connected to the motherboard via PCI bus and mounted at slot 2. This allows for running the Avanto and Espree systems as Tim [32 x 8] systems.
- Up to three additional optional PCI_RX boards (slot 3, slot 4 and slot 5) are connected to the motherboard via PCI bus. Three PCI_RX boards allow for running as Tim [76 x 18] systems. Four PCI_RX boards allow for running as a Tim [76 x 32] system.
- The Adaptec 2200 RAID controller which runs the four raw data disks as a RAID 0 system using SCSI U320 LVD is connected to the motherboard via a PCI bus. It is located at slot 1 and provides two external SCSI connectors (channel 0 and channel 1) which are not used.

Fig. 13: Imager MR005 Step2 Interface Boards - Overview



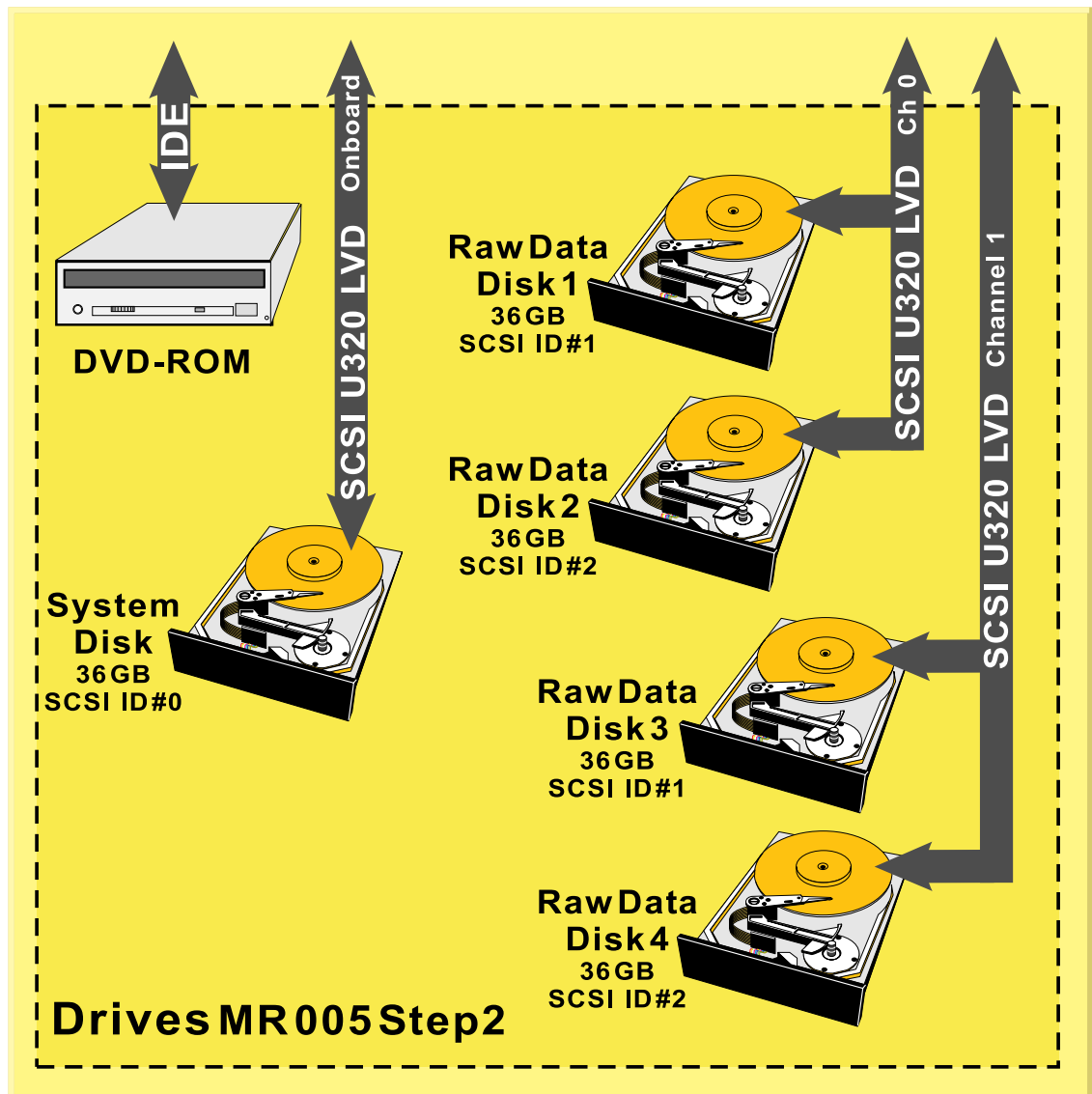
3.4 Drives of Imager MR 005 Step2 (Dual Opteron)

3.4.1 Function

There are six drives located at the Imager MR 005 Step2:

- The DVD-ROM drive is connected via an Integrated Device Equipment (IDE) interface. This internal DVD-ROM drive is used for software reload, software installation and software updates. For the DVD-ROM drive, the syngo MR software defines the logical drive name E:.
- The system hard disk drive is connected via an on-board SCSI bus system. The SCSI standard used here is a 16 bit standard called SCSI U 320 LVD. "U 320" means that it is an Ultra SCSI standard with a transfer rate of 320 MByte per second. LVD is an abbreviation for "Low Voltage Differential SCSI". The system hard disk drive uses the SCSI address or SCSI ID #0. It is a 36 GB HDD and the syngo MR software defines the logical drive name C:.
- There are four raw data disks connected via an Adaptec RAID controller. The four HDDs use two different SCSI channels: channel 0 and channel 1. The SCSI addresses or SCSI IDs used at each channel are ID #1 and ID #2. They are also 36 GB HDDs and used at the Imagers MR 005 Step2 to store raw data prior to starting the image calculation. For this particular RAID system with four raw data disks, the syngo MR software defines the logical drive name F:.

Fig. 14: Imager MR005 Step2 Drives - Overview



3.5 PCI_RX Boards at Imager MR 005 Step2 (Dual Opteron)

3.5.1 Function

The main task of the PCI_RX boards is to demodulate and filter the acquired digital MR signals coming from the receiver boards of the RFSU. Once processed, the resulting "raw data" are forwarded to the raw data hard drives for temporary storage.

The main components of the PCI_RX are :

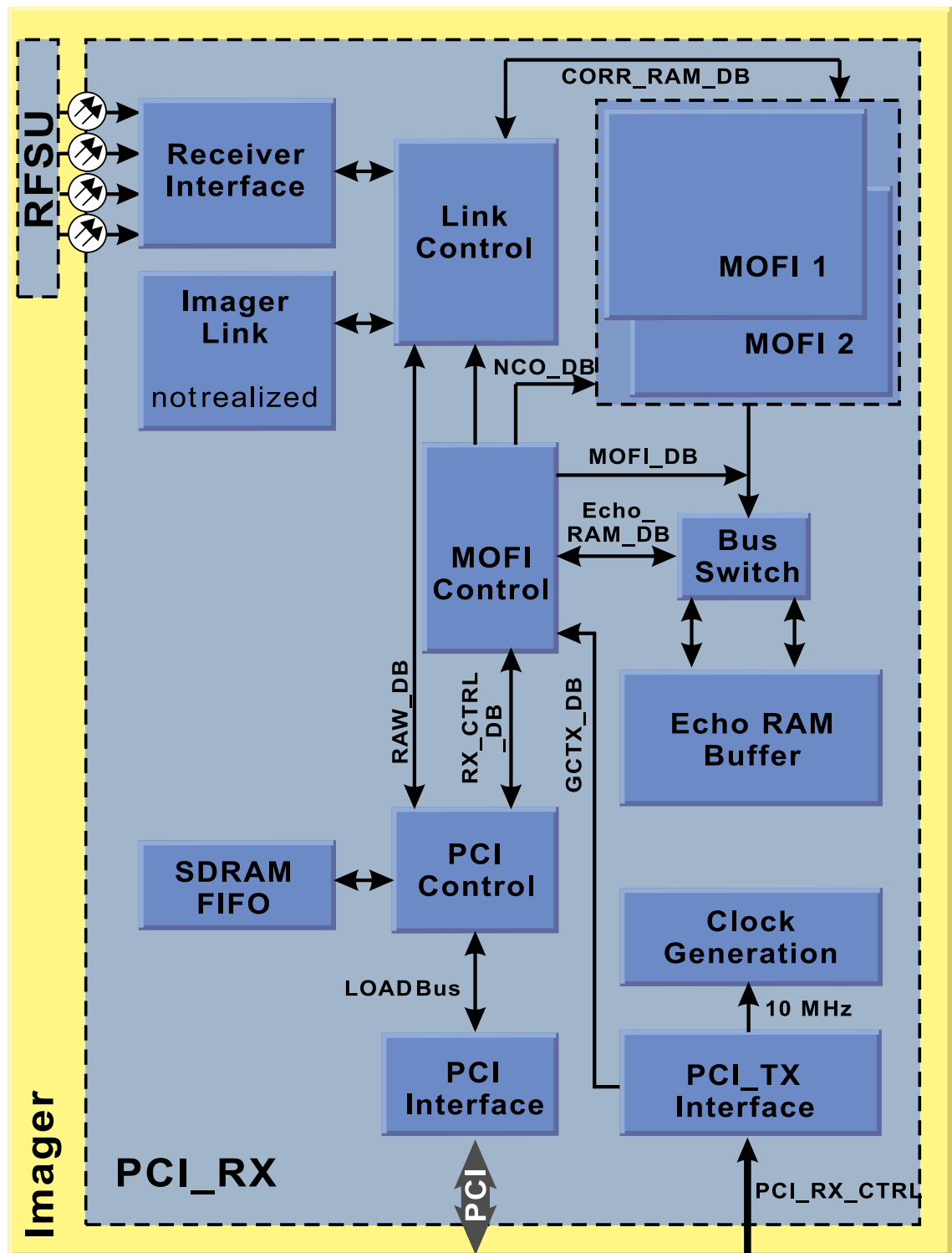
- PCI Interface,
- PCI Control,
- SDRAM FIFO as a buffer to the PCI bus,
- Receiver Interface with high speed FOC links,
- 2 MOFIs for demodulating and filtering four receive channels each,
- PCI_TX Interface to control the PCI_RX while running a sequence
- Echo RAM Buffer to store the MR echo signals

The following table provides you with an overview of the MRIR MR 005 Step2 configurations for Tim systems such as MAGNETOM Avanto, MAGNETOM Espree or SaTS:

Tab. 7 MRIR Configurations

Tim Configuration	# of PCI_RX boards	PCI slots used
Tim [32 x 8] (Basic)	1	2
Tim [76 x 18] (option)	3	2, 3, 4
Tim [76 x 32] (option)	4	2, 3, 4, 5

Fig. 15: Imager PCI_RX Overview



4.1 Overview of Imager MR 005 Step2 Standard (Dual Opteron)

The Imager MR 005 Step2 Standard (MRIR), also called MC Pro Multi 2, is a 19 inch rack-mounted industrial PC and is located in the ACC cabinet.

This Imager MR 005 Step2 Standard (MRIR) is used for all Tim systems (f. e. MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) systems since software syngo version MR B13.

The MR signal data coming from the RF system and demodulated as well as filtered at the PCI_RX16 will be temporarily stored on a raw data disk drive. The task of the Imager MR 005 Step2 Standard (MRIR) is to collect MR signals from the RFSU, to calculate the raw data, and to finally perform image reconstruction. The raw data are stored temporarily on the raw data disks. Two AMD processors perform image reconstruction after the measurement is finished. The resulting images are transferred via the internal MR network to the patient database of the MRC Host.

Raw data will be available for display only when enabling the storage prior to the measurement. The second possibility of displaying raw data is a backward FFT of the images.

The task of the Imager MR 005 Step2 Standard (MRIR) is:

- to receive the measurement data from the RF system
- to calculate the raw data during demodulation
- to reconstruct, to display, and to evaluate MR images

The next figure shows the 19 inch component of an Avanto Imager MR 005 Step2 Standard. It includes:

- the motherboard,
- five hard disk drives,
- several standard interfaces,
- and MR-specific interfaces.

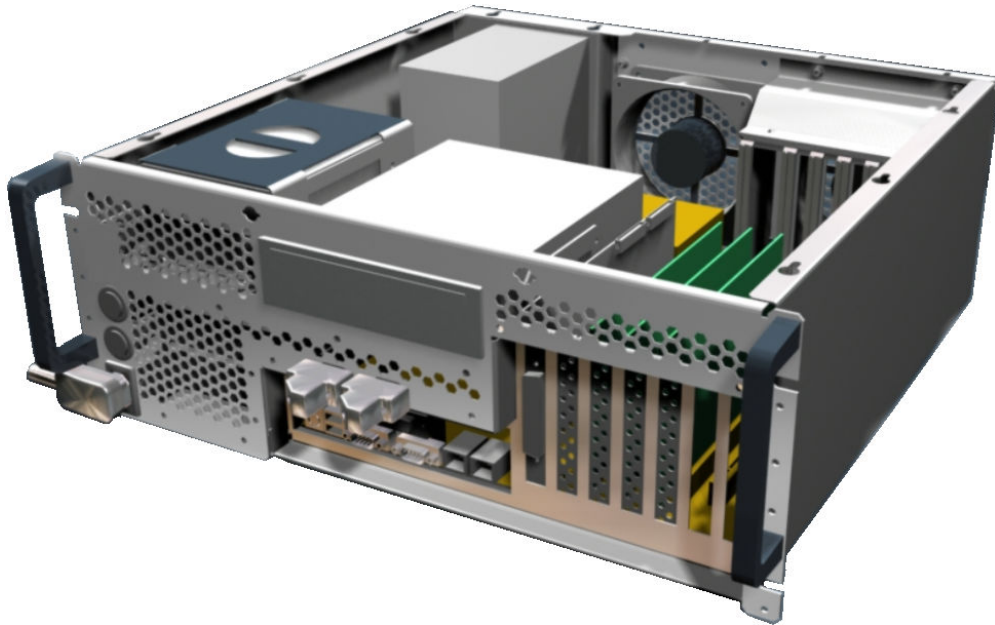
The details of the Imager MR 005 Step2 Standard are pretty similar to the details of the Imager MR 005 Step2 discussed before.

The main differences are :

- faster CPU speed (2.6 GHz versus 2.2 GHz)
- faster DRAM clocking (400 MHz versus 333 MHz)
- PCI_RX16 versus PCI_RX

However, due to rapid developments in the computer world, this description shows the conditions at the time of creation or modification of this document.

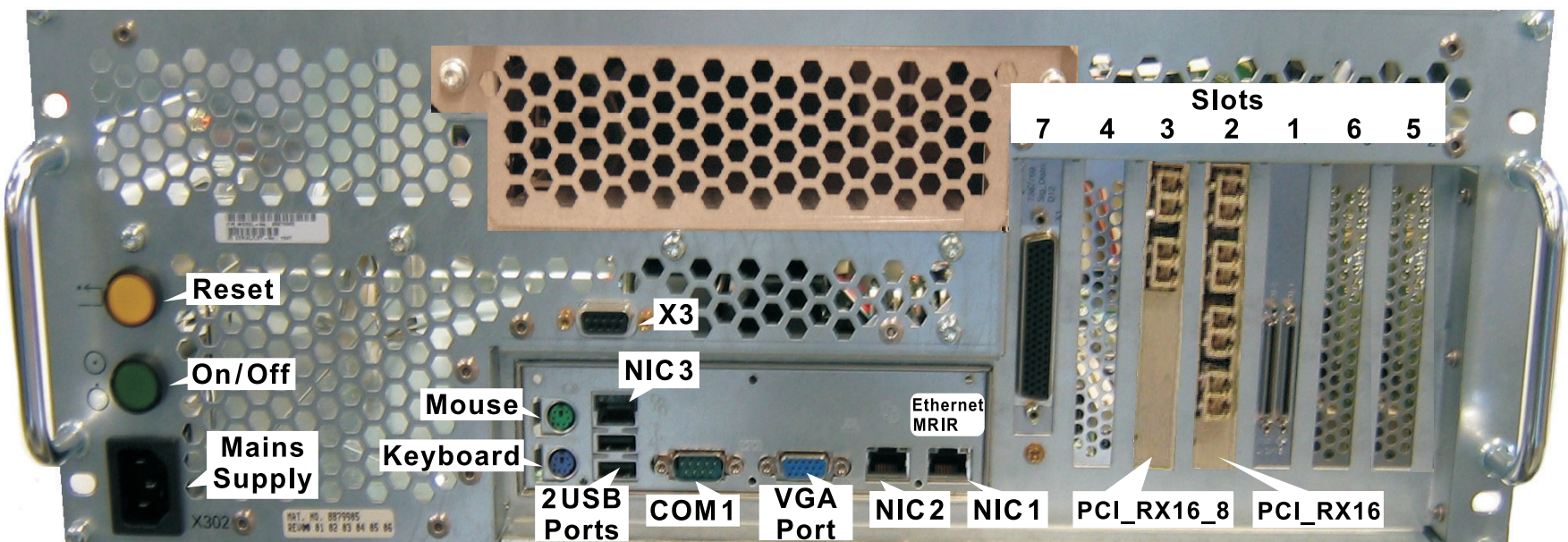
Fig. 16: Imager Unit - MR 005 Step2



A block diagram of the Imager MR 005 Step2 Standard systems used is shown in the following figure. The diagram shows the different areas on the Imager MR 005 Step2 Standard component, such as the motherboard, the drives, and the interface boards.

We have started to use a new software release, running Linux as operating system software on the Imager MR 005 Step2 Standard since syngo MR B13.

Fig. 17: Front View Imager MR005 Step2 Standard





Please take a look at the strange order of the slot numbers on the Imager MR 005 Step2 Standard.

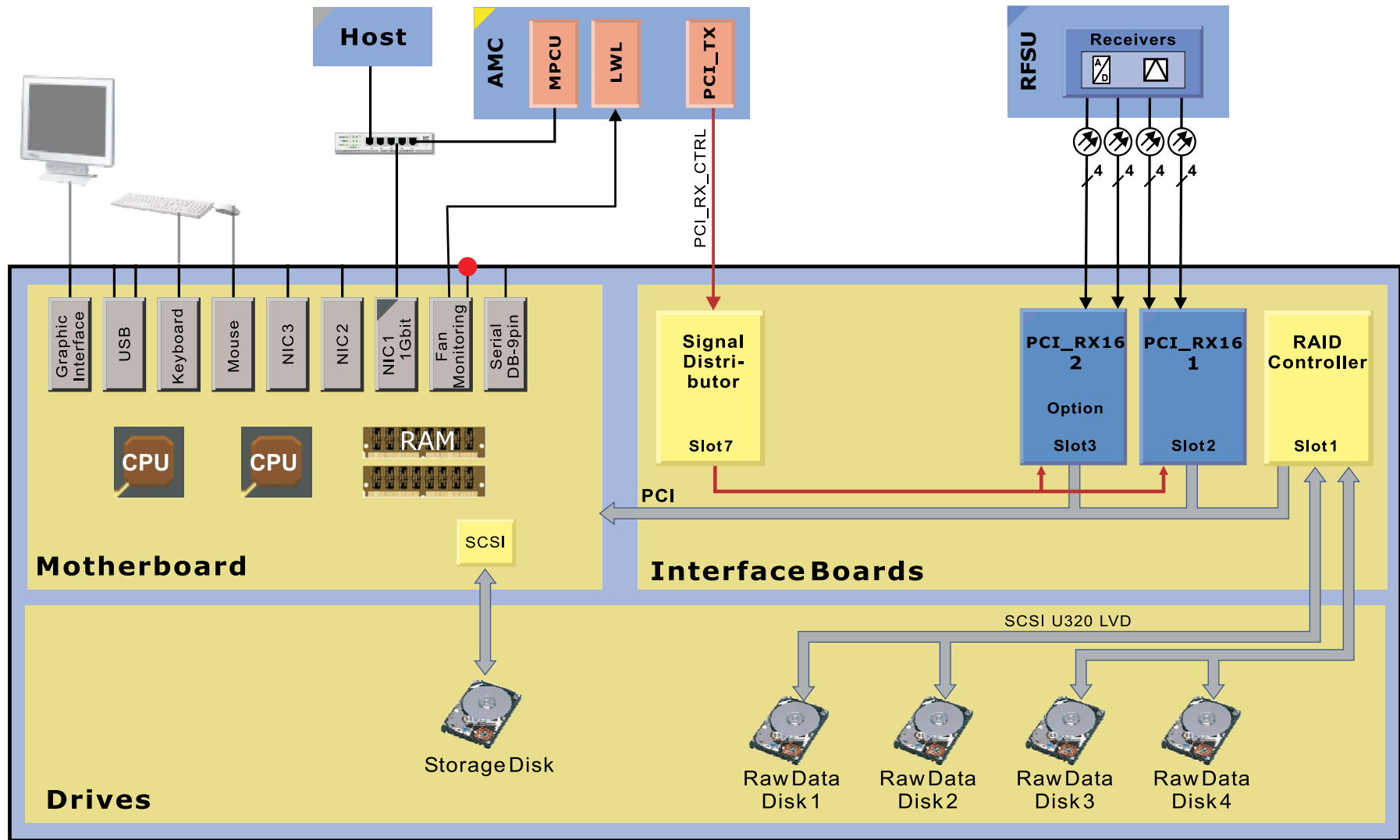
This order is correct!

Tab. 8 Imager MR005 Step 2 Standard Slots

Label	Use
Slot 5	not used
Slot 6	not used
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX16 board number 1
Slot 3	PCI_RX16 board number 2 (optional)
Slot 4	not used
Slot 7	Signal Distributor

Fig. 18: Imager MR005 Step2 Standard

Imager MR005 Step2 Standard



4.2 Motherboard of Imager MR 005 Step2 Standard (Dual Opteron)

4.2.1 Function

The Imager MR 005 Step2 Standard motherboard can be broken down into three functional blocks:

- CPU and RAM area
- on-board controllers
- peripheral interfaces (all accessible at the front side of the Imager MR 005 Step2 Standard)

4.2.1.1 CPU and RAM

The CPU and RAM area of the dual processor motherboard is equipped with:

- Dual 2.6 GHz AMD Opteron **CPUs** (64 bit CPUs) with 1MB second level cache
- 8 GB DDR SDRAM

4.2.1.2 On-board Controllers

- up to 2 PCI_RX16 boards
- 1x PCI SCSI controller card (Adaptec RAID-capable controller) for control of the four raw data scratch disks
- 1x onboard SCSI controller, an Adaptec AIC-7902 U320-SCSI-Controller for the system disk

4.2.1.3 Peripheral Interfaces

The peripheral interfaces include:

- 2x PS/2 ports, one keyboard, one mouse. The Imager MR 005 Step2 Standard is not equipped with a keyboard or a mouse or a keyboard + mouse emulator.
- 2x USB 1.1 ports - on the front side - not used
- Serial port (DB-9 pin) - not used
- On-board graphic controller (ATI Rage XL) - to connect a monitor, if necessary
- 2x ethernet adapters (Broadcom Gigabit) running at 1 Gigabit (NIC1, NIC2). The first ethernet adapter, NIC 1 (right connector), is used to connect the Imager MR 005 Step2 Standard to the Host and AMC.
- 1x ethernet adapter (Intel 10/100 Megabit), which is not used (NIC3).

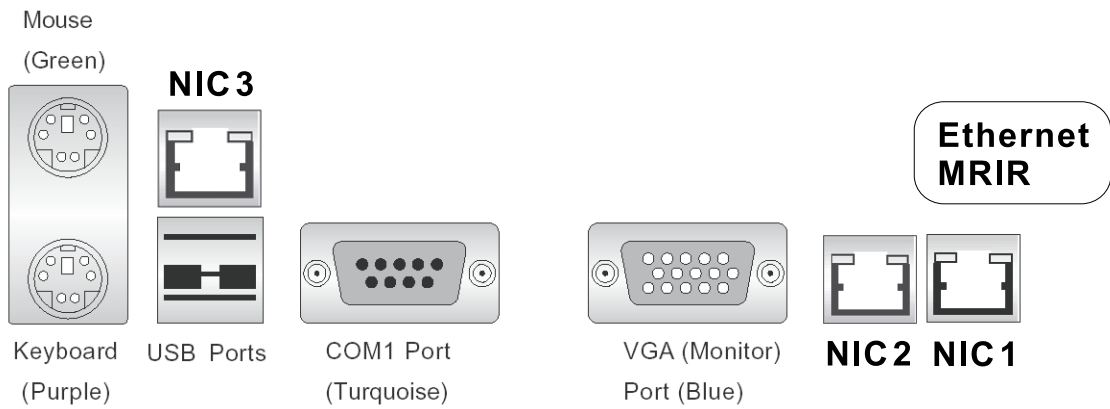
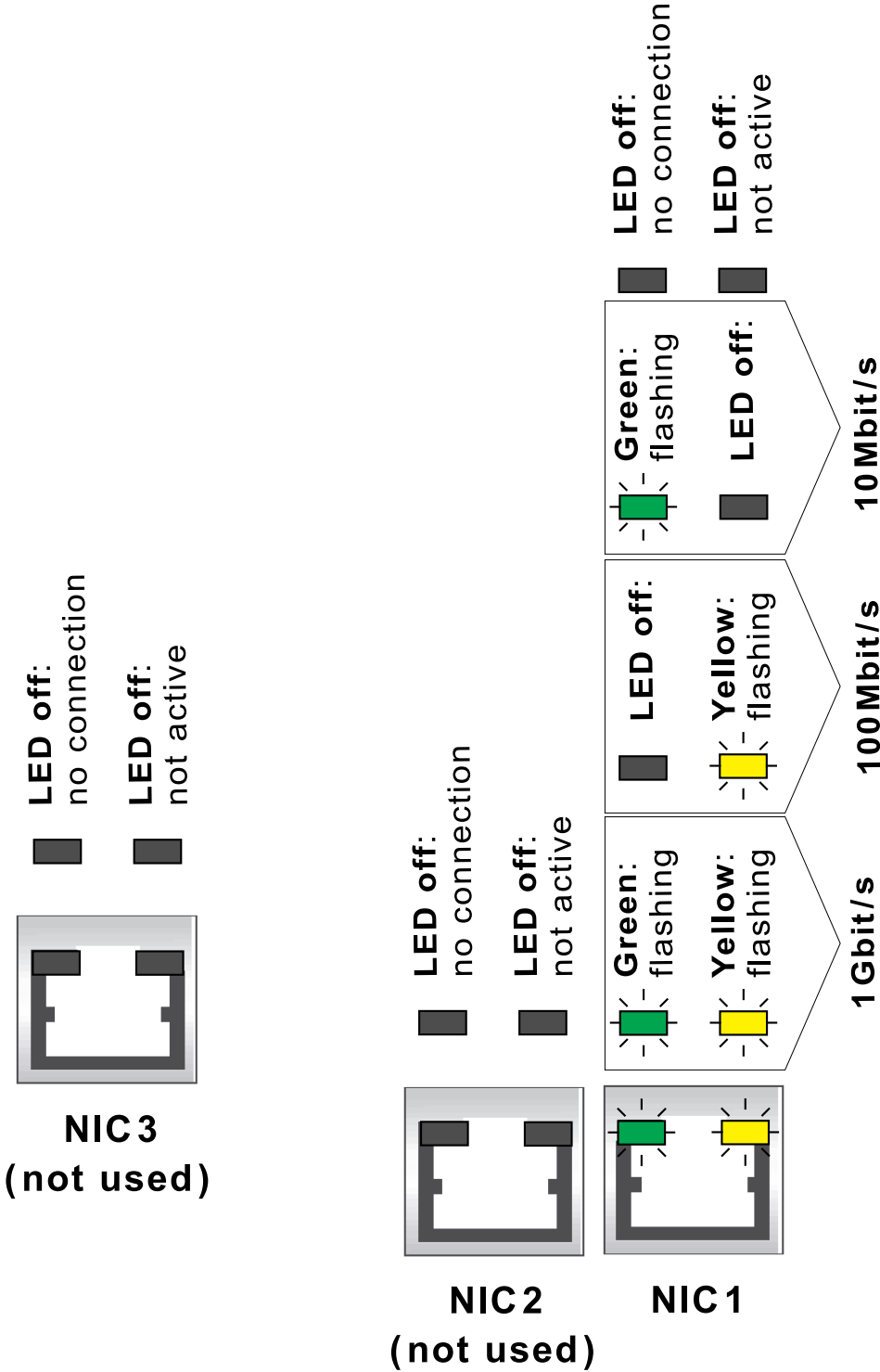
Fig. 19: Imager MR005 Step2 Peripheral Interfaces

Fig. 20: Imager MR005 Step2 NIC LEDs



4.3 Interface Boards of Imager MR 005 Step2 Standard (Dual Opteron)

4.3.1 Function

There are different interface boards located at the Imager MR 005 Step2 Standard. These interface boards can be accessed at the front side of the Imager MR 005 Step2 Standard at different slots:

- The signal distributor located at slot 7 of the Imager MR 005 Step2 Standard. It distributes the control signals from the PCI_TX in the AMC to all available PCI_RX16 boards.
- One PCI_RX (now PCI_RX16_8) board as a standard, connected to the motherboard via PCI bus and mounted at slot 2. This allows for running the Avanto and Espree systems as Tim [32 x 8] systems.
- Optional PCI_RX boards are:
One PCI_RX16 board (slot 2) together with one PCI_RX16_8 board (slot 3) connected to the motherboard via PCI bus for running as Tim [76 x 18] systems.
Or:
Two PCI_RX16 boards (slot 2 and slot 3) allow for running as a Tim [76 x 32] system.
- The Adaptec RAID controller which runs the four raw data disks as a RAID 0 system using SCSI U320 LVD is connected to the motherboard via a PCI bus. It is located at slot 1 and provides two external SCSI connectors (channel 0 and channel 1) which are not used.

4.4 Drives of Imager MR 005 Step2 Standard (Dual Opteron)

4.4.1 Function

There are five drives located at the Imager MR 005 Step2 Standard:

- The storage disk drive is connected via an on-board SCSI bus system. The SCSI standard used here is a 16 bit standard called SCSI U 320 LVD. "U 320" means that it is an Ultra SCSI standard with a maximum transfer rate of 320 MByte per second. LVD is an abbreviation for "Low Voltage Differential SCSI". The system hard disk drive uses the SCSI address or SCSI ID #0. It is a 36 GB HDD with two partitions and the logical drive names `/dev/sda1` and `/dev/sda2`.
- There are four raw data disks connected via an Adaptec RAID controller. The four HDDs use two different SCSI channels: channel 0 and channel 1. The SCSI addresses or SCSI IDs used at each channel are ID #1 and ID #2. They are also 36 GB HDDs and used at the Imagers MR 005 Step2 Standard to store raw data prior to starting the image calculation. For this particular RAID system with four raw data disks the software defines the logical drive name `/dev/sdb1`.

4.5 PCI_RX16 Boards at Imager MR 005 Step2 Standard (Standard Opteron)

4.5.1 Function

The main task of the PCI_RX (PCI_RX16 or/and PCI_RX16_8) boards is to demodulate and filter the acquired digital MR signals coming from the receiver boards of the RFSU. Once processed, the resulting "raw data" are forwarded to the raw data hard drives for temporary storage.

The main components of the PCI_RX16 are :

- PCI Interface,
- PCI Control,
- SDRAM FIFO as a buffer to the PCI bus,
- Receiver Interface with high speed FOC links,
- MOFIs for demodulating and filtering eight receive channels each,
- PCI_TX Interface to control the PCI_RX16 while running a sequence
- Echo RAM Buffer to store the MR echo signals

The following table provides you with an overview of the MRIR MR 005 Step2 Standard configurations for Tim systems such as for example MAGNETOM Avanto, MAGNETOM Es-
preo or SaTS and TaTS:

Tab. 9 MRIR Configurations

Tim Configuration	# of PCI_RX boards	PCI slots used
Tim [32 x 8] (Basic)	1 PCI_RX16_8	2
Tim [76 x 18] (option)	1 PCI_RX16 and 1 PCI_RX16_8	2, 3
Tim [76 x 32] (option)	2 PCI_RX16	2, 3

PCI_RX16 means a PCI_RX board able to handle 16 RF receive channels.

PCI_RX16_8 means a PCI_RX board able to handle 8 instead of 16 RF receive channels.

5.1 Overview of Imager MR 005 Step2 High End (Quad Opteron)

The Imager MR 005 Step2 High End (MRIR), also called MC Pro Multi 8, is a 19 inch rack-mounted industrial PC is located in the ACC cabinet. It is realized in two boxes, the Base Unit and the Disc Drawer.

This optional Imager MR 005 Step2 High End (MRIR) is available for all Tim systems (f. e. MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) since software syngo MR B13.

The MR signal data coming from the RF system and demodulated as well as filtered at the PCI_RX will be temporarily stored a raw data drive. The task of the Imager MR 005 Step2 High End (MRIR) is to collect MR signals from the RFSU, to calculate the raw data, and to finally perform image reconstruction. The raw data are stored temporarily on the raw data disks. Four dual core AMD processors perform image reconstruction after the measurement is finished. The resulting images are transferred via the internal MR network to the patient database of the MRC Host.

Raw data will be available for display only when enabling the storage prior to the measurement. The second possibility of displaying raw data is a backward FFT of the images.

The task of the Imager MR 005 Step2 High End (MRIR) is:

- to receive the measurement data from the RF system
- to calculate the raw data during demodulation
- to reconstruct, to display, and to evaluate MR images

The next figure shows the 19 inch component of an Avanto Imager MR 005 Step2 High End. It includes:

- the motherboard,
- five hard disk drives,
- several standard interfaces,
- and MR-specific interfaces.

However, due to rapid developments in the computer world, this description shows the conditions at the time of creation or modification of this document.

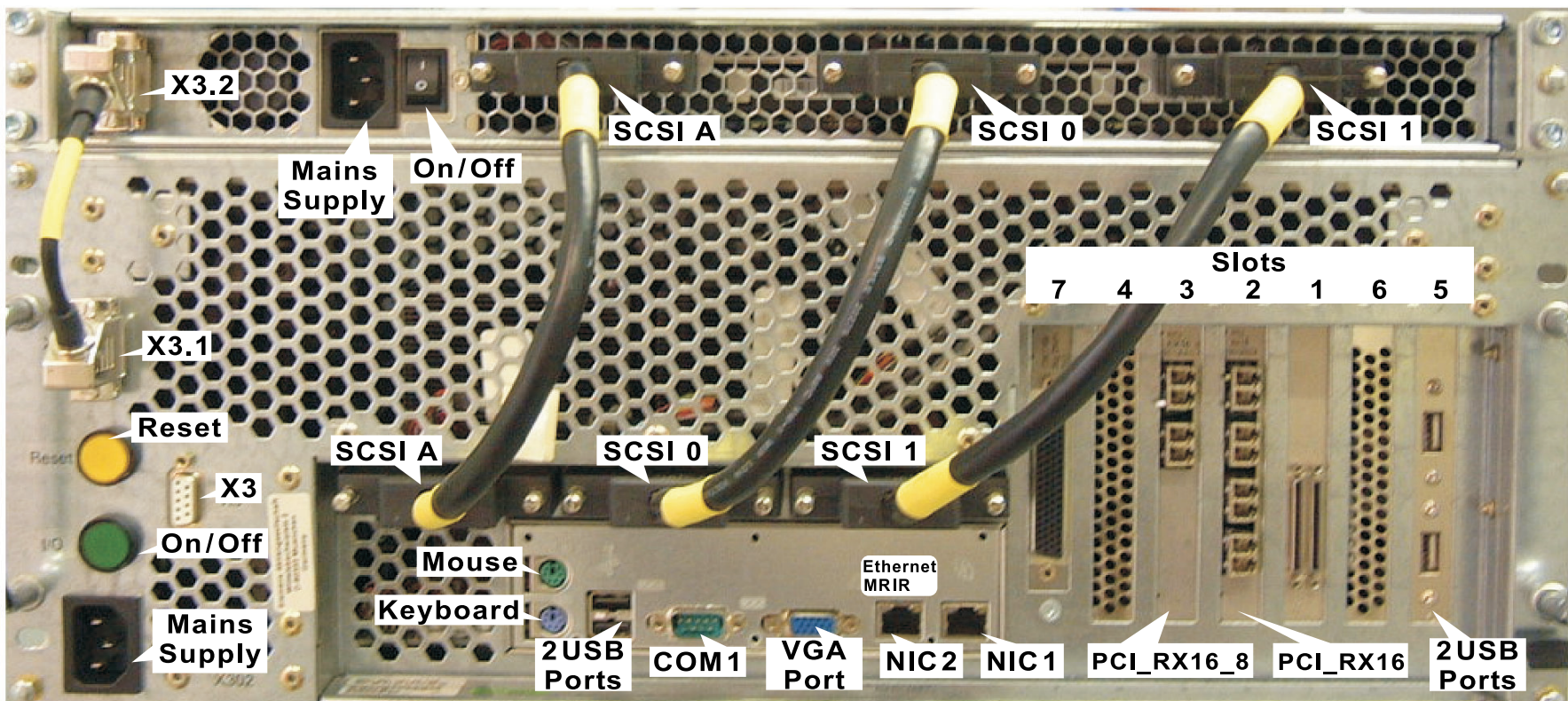
Fig. 21: Imager Unit MR005 Step2 High End



A block diagram of the Imager MR 005 Step2 High End systems used is shown in the following figure. The diagram shows the different areas on the Imager MR 005 Step2 High End component, such as the motherboard, the drives, and the interface boards.

This optional imager is supported since syngo MR B13. Here we run Linux as operation system software.

Fig. 22: Front View Imager MR005 Step2 High End





Please take a look at the strange order of the slot numbers on the Imager MR 005 Step2 High End.

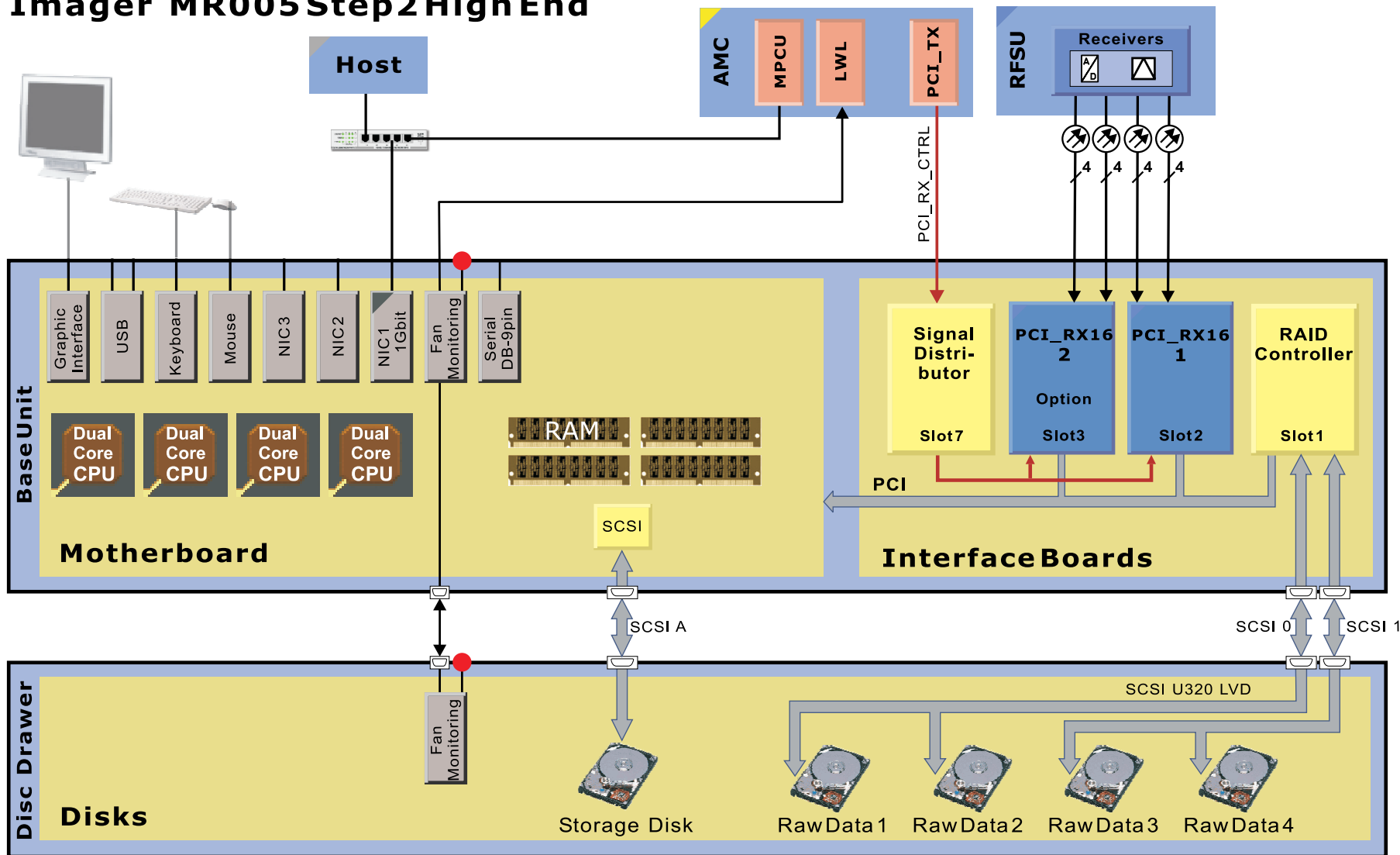
This order is correct!

Tab. 10 Imager MR005 Step2 High End Slots

Label	Use
Slot 5	not used
Slot 6	not used
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX16 board number 1
Slot 3	PCI_RX16 board number 2 (optional)
Slot 4	not used
Slot 7	Signal Distributor

Fig. 23: Imager MR005 Step2 High End

Imager MR005 Step2 High End



5.2 Motherboard of Imager MR 005 Step2 High End (Quad Opteron)

5.2.1 Function

The Imager MR 005 Step2 High End motherboard can be broken down into three functional blocks:

- CPU and RAM area
- on-board controllers
- peripheral interfaces (all accessible at the front side of the Imager MR 005 Step2 High End)

5.2.1.1 CPU and RAM

The CPU and RAM area of the quad processor motherboard is equipped with:

- Dual Core 2.2 GHz AMD Opteron **CPUs** (64 bit CPUs) with 2MB second level cache
- 16 GB DDR SDRAM

5.2.1.2 On-board Controllers

- One or two PCI_RX boards
- 1x PCI SCSI controller card (Adaptec RAID-capable controller) for control of the four raw data scratch disk
- 1x onboard SCSI controller, an Adaptec AIC-7902 U320-SCSI-Controller for the system disk

5.2.1.3 Peripheral Interfaces

The peripheral interfaces include:

- 2x PS/2 ports, one keyboard, one mouse. The Imager MR 005 Step2 High End is not equipped with a keyboard or a mouse or a keyboard + mouse emulator.
- 2x USB ports - on the front side - not used
- 2x USB ports - put in at slot 5 - not used
- Serial port (DB-9 pin) - not used
- On-board graphic controller (ATI Rage XL) - to connect a monitor, if necessary
- 2x ethernet adapters (Broadcom Gigabit) running at 1 Gigabit (NIC1, NIC2). The first ethernet adapter, NIC 2 (left connector), is used to connect the Imager MR 005 Step2 High End to the Host and AMC.

Fig. 24: Imager MR005 Step2 High End Periperal Interfaces

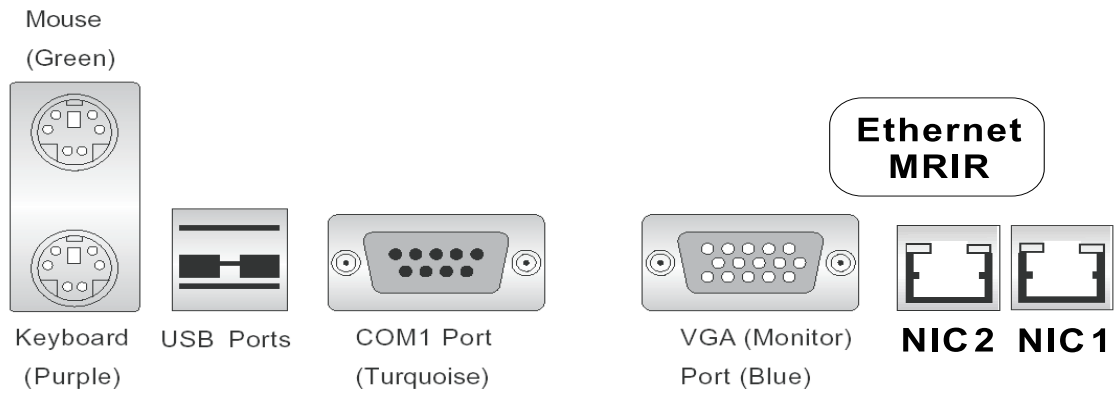
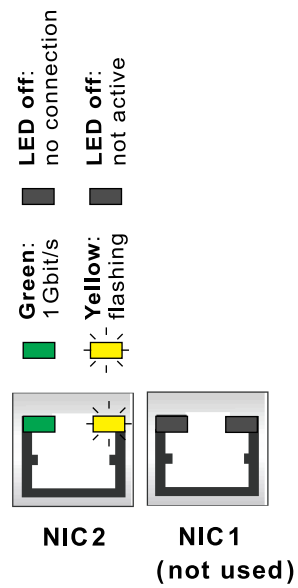


Fig. 25: Imager MR005 Step2 High End



5.3 Interface Boards of Imager MR 005 Step2 High End (Quad Opteron)

5.3.1 Function

There are different interface boards located at the Imager MR 005 Step2 High End. These interface boards can be accessed at the front side of the Imager MR 005 Step2 High End at different slots:

- The signal distributor located at slot 7 of the Imager MR 005 Step2 High End. It distributes the control signals from the PCI_TX in the AMC to the available PCI_RX16 boards.
- One PCI_RX16 (now PCI_RX16_8) board as a standard, connected to the motherboard via PCI bus and mounted at slot 2. This allows for running the Avanto and Espree systems as Tim [32 x 8] systems.
- Optional PCI_RX16 boards are:
One PCI_RX16 board (slot 2) together with one PCI_RX16_8 board (slot 3) connected to the motherboard via PCI bus for running as Tim [76 x 18] systems.
Or:
Two PCI_RX16 boards (slot 2 and slot 3) allow for running as a Tim [76 x 32] system.
- The Adaptec RAID controller which runs the four raw data disks as a RAID 0 system using SCSI U320 LVD is connected to the motherboard via a PCI bus. It is located at slot 1 and provides two external SCSI connectors (channel 0 and channel 1) which are not used.

5.4 Drives of Imager MR 005 Step2 High End (Quad Opteron)

5.4.1 Function

There are six drives located at the Imager MR 005 Step2 High End:

- The storage disk drive is connected via an on-board SCSI bus system. The SCSI standard used here is a 16 bit standard called SCSI U 320 LVD. "U 320" means that it is an Ultra SCSI standard with a maximum transfer rate of 320 MByte per second. LVD is an abbreviation for "Low Voltage Differential SCSI". The storage disk drive uses the SCSI address or SCSI ID #0. It is a 73 GB HDD and the software defines the logical drive names `/dev/sda1` and `/dev/sda2`.
- There are four raw data disks connected via an Adaptec RAID controller. The four HDDs use two different SCSI channels: channel 0 and channel 1. The SCSI addresses or SCSI IDs used at each channel are ID #1 and ID #2. They are also 73 GB HDDs and used at the Imagers MR 005 Step2 High End to store raw data prior to starting the image calculation. For this particular RAID system with four raw data disks, the software defines the logical drive name `/dev/sdb1`.

5.5 PCI_RX16 Boards at Imager MR 005 Step2 High End (Quad Opteron)

5.5.1 Function

The main task of the PCI_RX16 boards is to demodulate and filter the acquired digital MR signals coming from the receiver boards of the RFSU. Once processed, the resulting "raw data" are forwarded to the raw data hard drives for temporary storage.

The main components of the PCI_RX16 are :

- PCI Interface,
- PCI Control,
- SDRAM FIFO as a buffer to the PCI bus,
- Receiver Interface with high speed FOC links,
- 2 MOFIs for demodulating and filtering eight receive channels each,
- PCI_TX Interface to control the PCI_RX16 while running a sequence
- Echo RAM Buffer to store the MR echo signals

The following table provides you with an overview of the MRIR MR 005 Step2 High End configurations for Tim systems such as MAGNETOM Avanto, MAGNETOM Espree or SaTS and TaTS:

Tab. 11 MRIR Configurations

Tim Configuration	# of PCI_RX boards	PCI slots used
Tim [32 x 8] (Basic)	1 PCI_RX16_8	2
Tim [76 x 18] (option)	1 PCI_RX16 and 1 PCI_RX16_8	2, 3
Tim [76 x 32] (option)	2 PCI_RX16	2, 3

PCI_RX16 means a PCI_RX board able to handle 16 RF receive channels.

PCI_RX16_8 means a PCI_RX board able to handle 8 instead of 16 RF receive channels.

6.1 Overview of Imager MR 005 Step3 Standard

The Imager MR 005 Step3 Standard (MRIR), also called MC Pro Multi 4, is a 19 inch rack-mounted industrial PC and is located in the ACC cabinet.

This Imager MR 005 Step3 Standard (MRIR) is used for all Tim systems (f. e. MAGNETOM Avanto, MAGNETOM Espree, SaTS and TaTS) systems since software version syngo MR B13.

This Imager MR 005 Step3 Standard is very similar to the former discussed Imager MR 005 Step2 Standard especially from the interfaces point of view.

The MR signal data coming from the RF system and demodulated as well as filtered at the PCI_RX16 will be temporarily stored on a raw data disk drive. The task of the Imager MR 005 Step3 Standard (MRIR) is to collect MR signals from the RFSU, to calculate the raw data, and to finally perform image reconstruction. The raw data are stored temporarily on the raw data disks. Two AMD Dual Core processors perform image reconstruction after the measurement is finished. The resulting images are transferred via the internal MR network to the patient database of the MRC Host.

Raw data will be available for display only when enabling the storage prior to the measurement. The second possibility of displaying raw data is a backward FFT of the images.

The task of the Imager MR 005 Step3 Standard (MRIR) is:

- to receive the measurement data from the RF system
- to calculate the raw data during demodulation
- to reconstruct, to display, and to evaluate MR images

The Avanto Imager MR 005 Step3 Standard includes:

- the motherboard,
- five hard disk drives,
- several standard interfaces,
- and MR-specific interfaces.

The details of the Imager MR 005 Step3 Standard are pretty similar to the details of the Imager MR 005 Step2 Standard discussed before.

However, due to rapid developments in the computer world, this description shows the conditions at the time of creation or modification of this document.

We have started running Linux as operating system software on the Imager MR 005 Step3 Standard since syngo MR B13.

Tab. 12 Imager MR005 Step 2 Standard Slots

Label	Use
Slot 5	not used
Slot 6	not used
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX16 board number 1
Slot 3	PCI_RX16 board number 2 (optional)

Label	Use
Slot 4	not used
Slot 7	Signal Distributor

6.2 Motherboard of Imager MR 005 Step3 Standard

6.2.1 Function

The Imager MR 005 Step3 Standard motherboard can be broken down into three functional blocks:

- CPU and RAM area
- on-board controllers
- peripheral interfaces (all accessible at the front side of the Imager MR 005 Step3 Standard)

6.2.1.1 CPU and RAM

The CPU and RAM area of the dual processor motherboard is equipped with:

- 2x Dual Core 2.4 GHz AMD-Socket-F-Opteron **CPUs** (64 bit CPUs) with 2x 1MB second level cache
- 8 GB or 16 GB RAM

6.2.1.2 On-board Controllers

- up to 2 PCI_RX16 boards
- 1x PCI SCSI controller card (Adaptec RAID-capable controller) for control of the four raw data scratch disks
- 1x onboard SCSI controller for the storage disk

6.2.1.3 Peripheral Interfaces

The peripheral interfaces include:

- 2x PS/2 ports, one keyboard, one mouse. The Imager MR 005 Step3 Standard is not equipped with a keyboard or a mouse or a keyboard + mouse emulator.
- 2x USB 1.1 ports - on the front side - not used
- Serial port (DB-9 pin) - not used
- On-board graphic controller (ATI Rage XL) - to connect a monitor, if necessary
- 2x ethernet adapters running at 1 Gigabit (NIC1, NIC2). The first ethernet adapter, NIC 1 (right connector), is used to connect the Imager MR 005 Step3 Standard to the Host and AMC.
- 1x ethernet adapter (Intel 10/100 Megabit), which is not used (NIC3).

6.3 Interface Boards of Imager MR 005 Step3 Standard

6.3.1 Function

There are different interface boards located at the Imager MR 005 Step3 Standard. These interface boards can be accessed at the front side of the Imager MR 005 Step3 Standard at different slots:

- The signal distributor located at slot 7 of the Imager MR 005 Step3 Standard. It distributes the control signals from the PCI_TX in the AMC to all available PCI_RX16 boards.
- One PCI_RX (now PCI_RX16_8) board as a standard, connected to the motherboard via PCI bus and mounted at slot 2. This allows for running the Avanto and Espree systems as Tim [32 x 8] systems.
- Optional PCI_RX boards are:

One PCI_RX16 board (slot 2) together with one PCI_RX16_8 board (slot 3) connected to the motherboard via PCI bus for running as Tim [76 x 18] systems.

Or:

Two PCI_RX16 boards (slot 2 and slot 3) allow for running as a Tim [76 x 32] system.

- The Adaptec RAID controller which runs the four raw data disks as a RAID 0 system using SCSI U320 LVD is connected to the motherboard via a PCI bus. It is located at slot 1 and provides two external SCSI connectors (channel 0 and channel 1) which are not used.

6.4 Drives of Imager MR 005 Step3 Standard

6.4.1 Function

There are five drives located at the Imager MR 005 Step3 Standard:

- The storage disk drive is connected via an on-board SCSI bus system. The SCSI standard used here is a 16 bit standard called SCSI U 320 LVD. "U 320" means that it is an Ultra SCSI standard with a maximum transfer rate of 320 MByte per second. LVD is an abbreviation for "Low Voltage Differential SCSI". The system hard disk drive uses the SCSI address or SCSI ID #0. It is a 36 GB HDD with two partitions and the logical drive names `/dev/sda1` and `/dev/sda2`.
- There are four raw data disks connected via an Adaptec RAID controller. The four HDDs use two different SCSI channels: channel 0 and channel 1. The SCSI addresses or SCSI IDs used at each channel are ID #1 and ID #2. They are also 36 GB HDDs and used at the Imagers MR 005 Step3 Standard to store raw data prior to starting the image calculation. For this particular RAID system with four raw data disks the software defines the logical drive name `/dev/sdb1`.

6.5 PCI_RX16 Boards at Imager MR 005 Step3 Standard

6.5.1 Function

The main task of the PCI_RX (PCI_RX16 or/and PCI_RX16_8) boards is to demodulate and filter the acquired digital MR signals coming from the receiver boards of the RFSU. Once processed, the resulting "raw data" are forwarded to the raw data hard drives for temporary storage.

The main components of the PCI_RX16 are :

- PCI Interface,
- PCI Control,
- SDRAM FIFO as a buffer to the PCI bus,
- Receiver Interface with high speed FOC links,
- MOFIs for demodulating and filtering eight receive channels each,
- PCI_TX Interface to control the PCI_RX16 while running a sequence
- Echo RAM Buffer to store the MR echo signals

The following table provides you with an overview of the MRIR MR 005 Step3 Standard configurations for Tim systems such as for example MAGNETOM Avanto, MAGNETOM Es-
pre or SaTS and TaTS:

Tab. 13 MRIR Configurations

Tim Configuration	# of PCI_RX boards	PCI slots used
Tim [32 x 8] (Basic)	1 PCI_RX16_8	2
Tim [76 x 18] (option)	1 PCI_RX16 and 1 PCI_RX16_8	2, 3
Tim [76 x 32] (option)	2 PCI_RX16	2, 3

PCI_RX16 means a PCI_RX board able to handle 16 RF receive channels.

PCI_RX16_8 means a PCI_RX board able to handle 8 instead of 16 RF receive channels.

- » The Host CELSIUS R640 and the Imager MR 005 Step3 Standard are included now.

There are no Hazard IDs in this document.

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Print No.: M6-015.850.12.02.02 | Replaces: M6-015.850.12.01.02
Doc. Gen. Date: 05.07 | Language: English
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